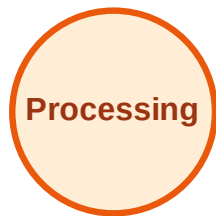


# BFS Traversal

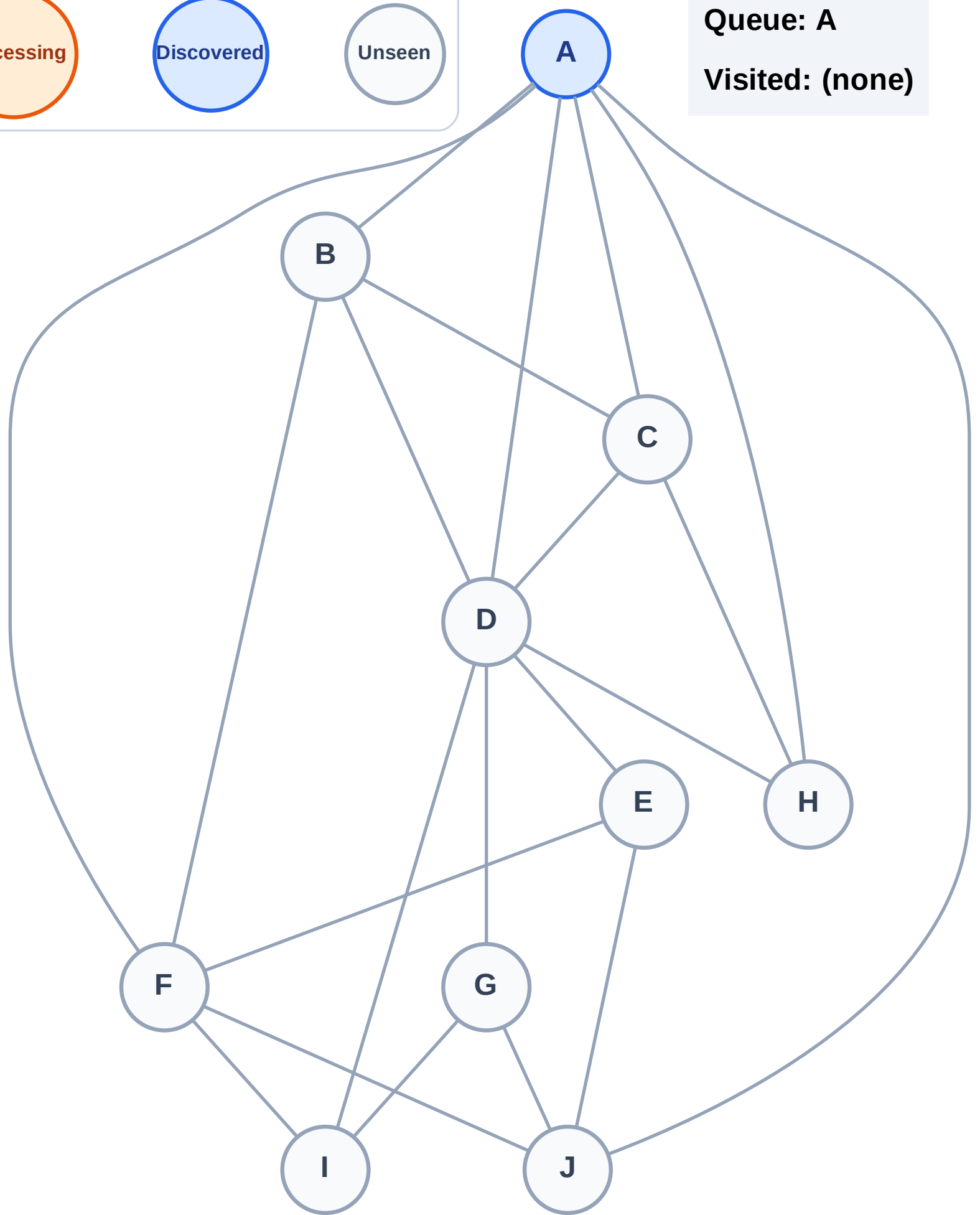
Step 1: Start at A

## Legend



Queue: A

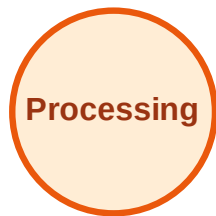
Visited: (none)



# BFS Traversal

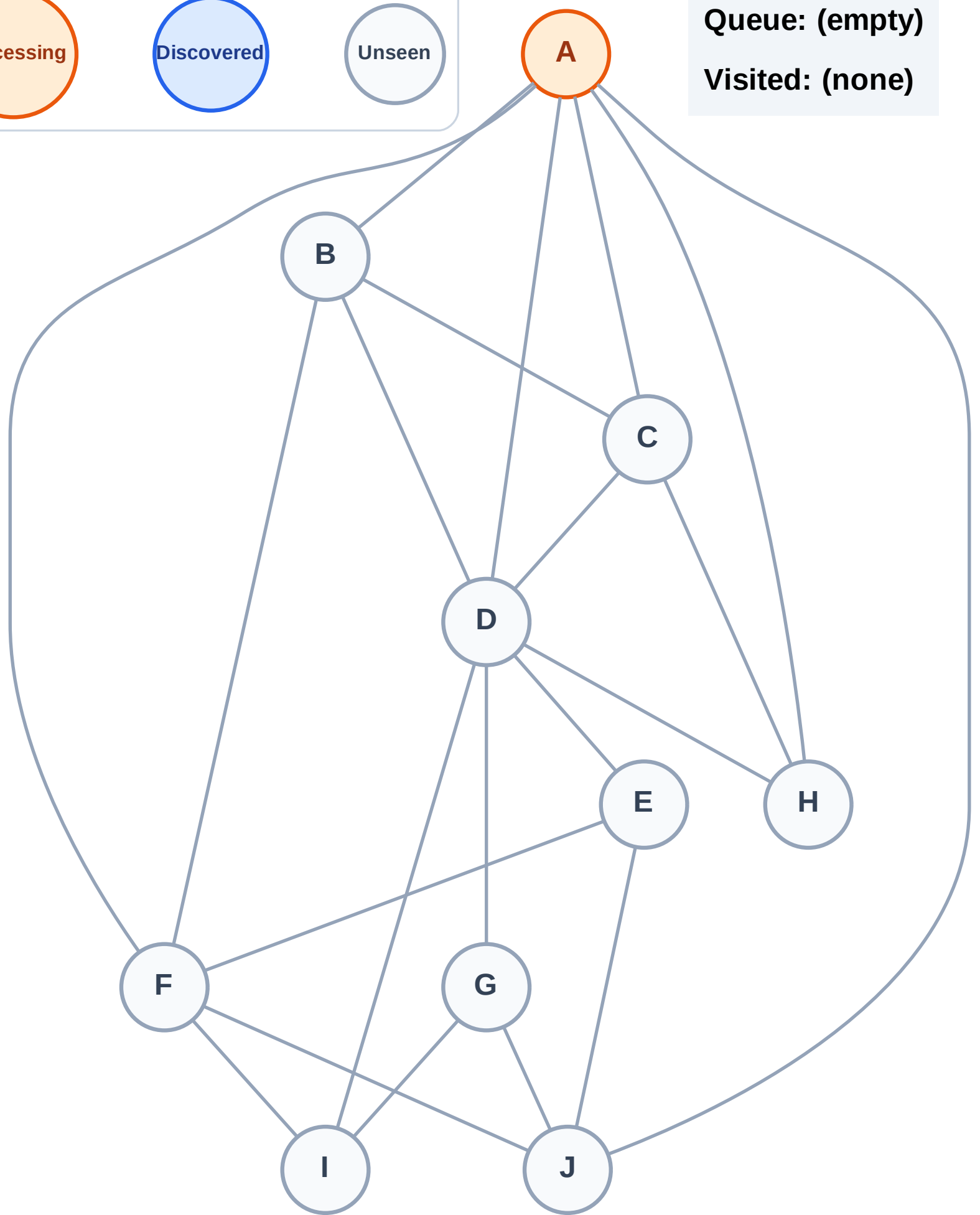
Step 2: Process node A

## Legend



Queue: (empty)

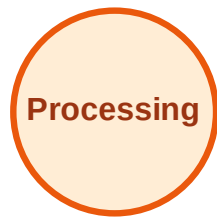
Visited: (none)



# BFS Traversal

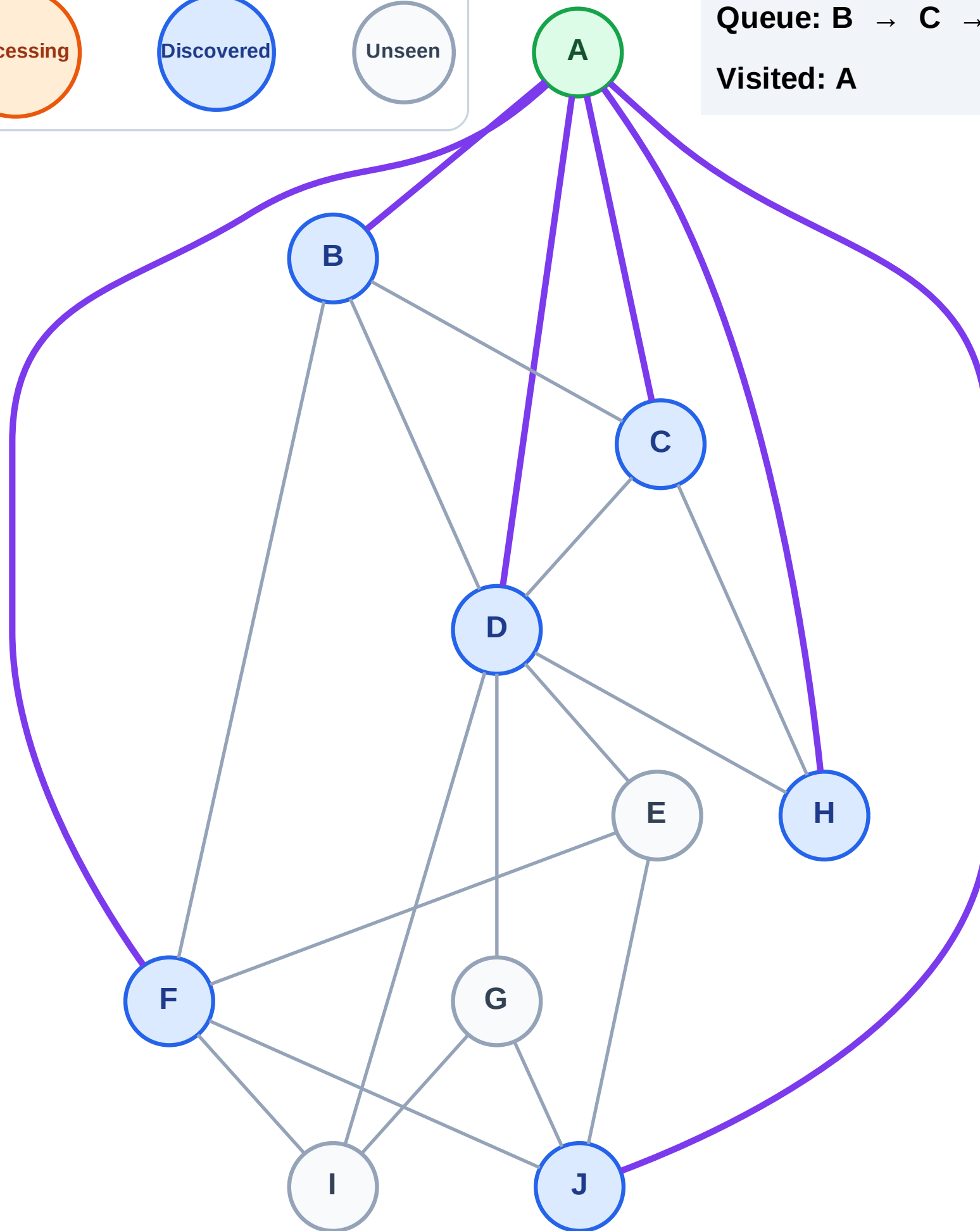
Step 3: Discover B, C, D, F, H, J from A

## Legend



Queue: B → C → D → F → H → J

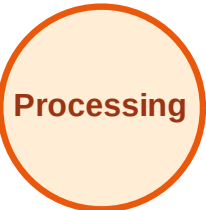
Visited: A



# BFS Traversal

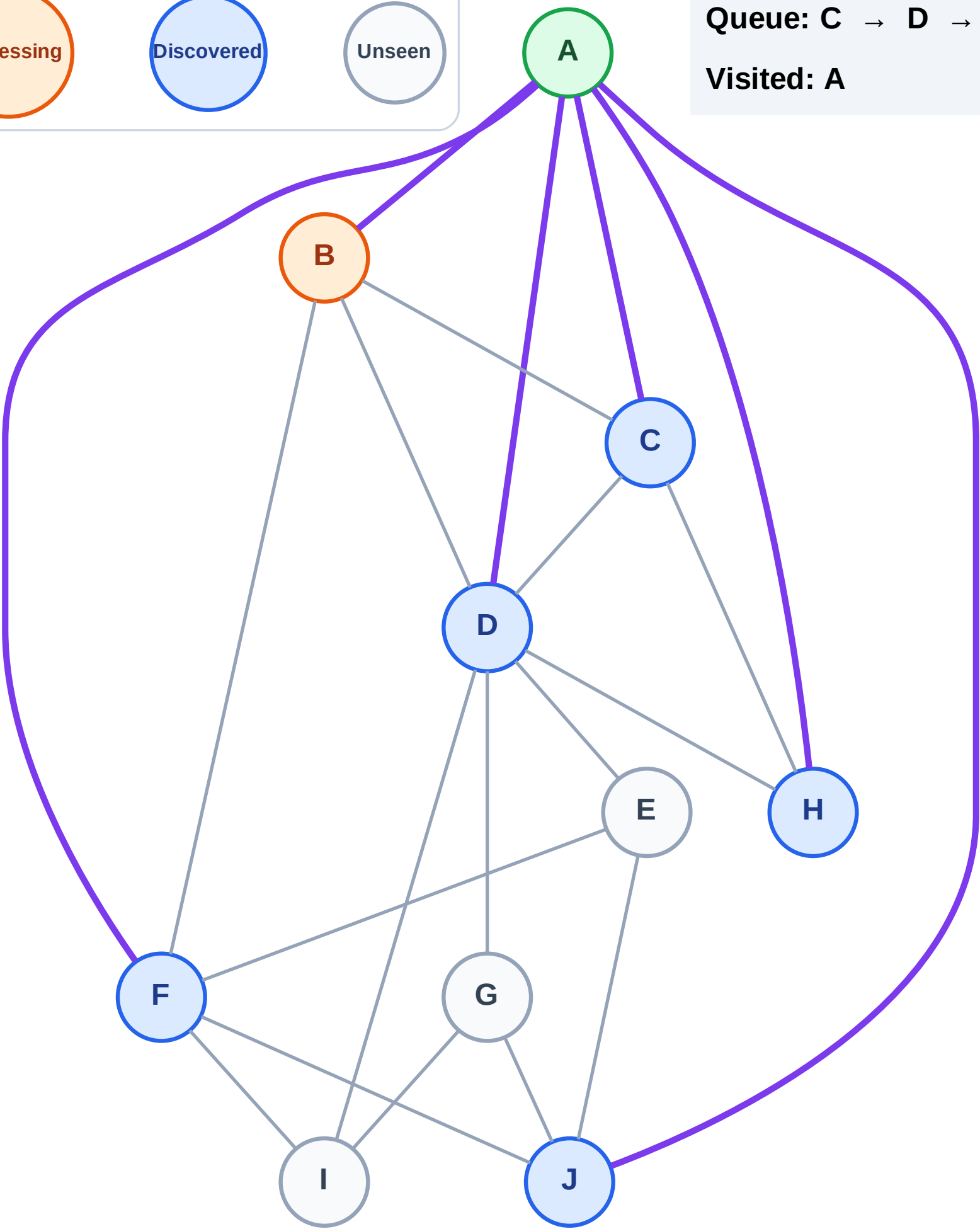
Step 4: Process node B

## Legend



Queue: C → D → F → H → J

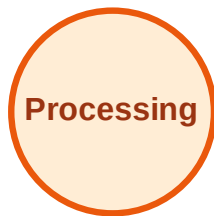
Visited: A



# BFS Traversal

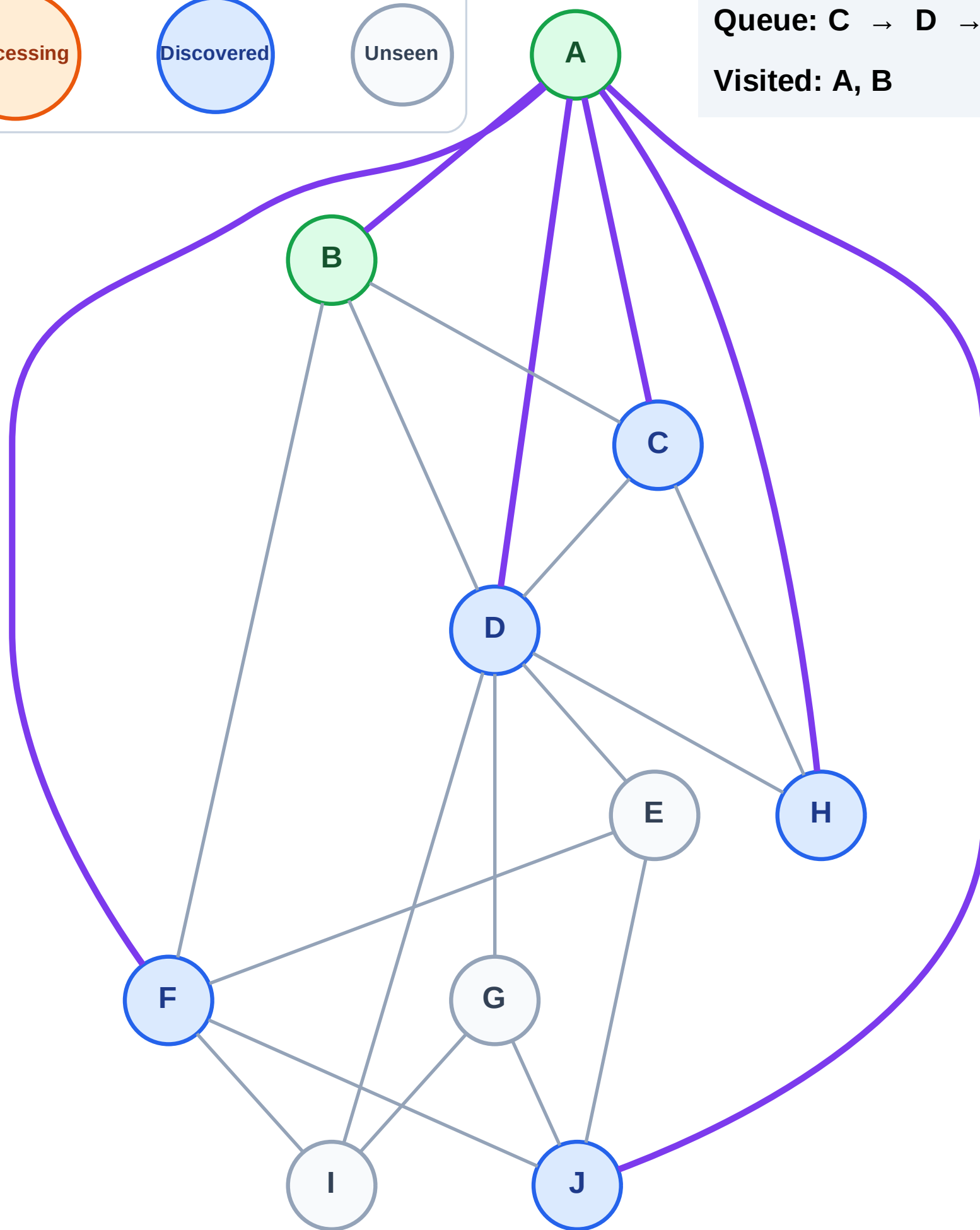
Step 5: No new neighbors from B

## Legend



Queue: C → D → F → H → J

Visited: A, B



# BFS Traversal

Step 6: Process node C

Legend

Complete

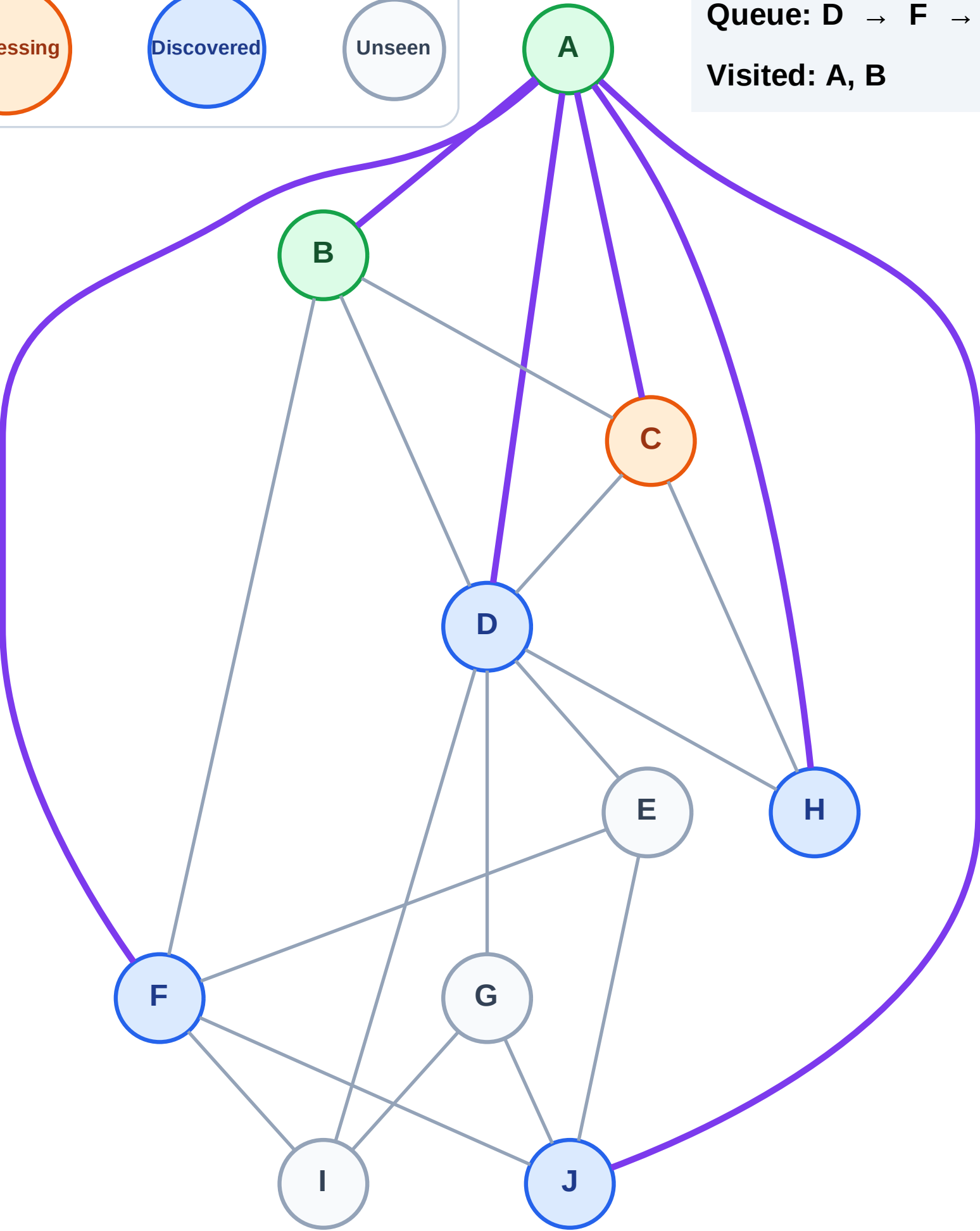
Processing

Discovered

Unseen

Queue: D → F → H → J

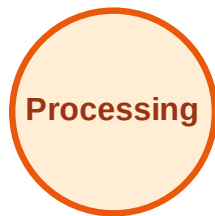
Visited: A, B



# BFS Traversal

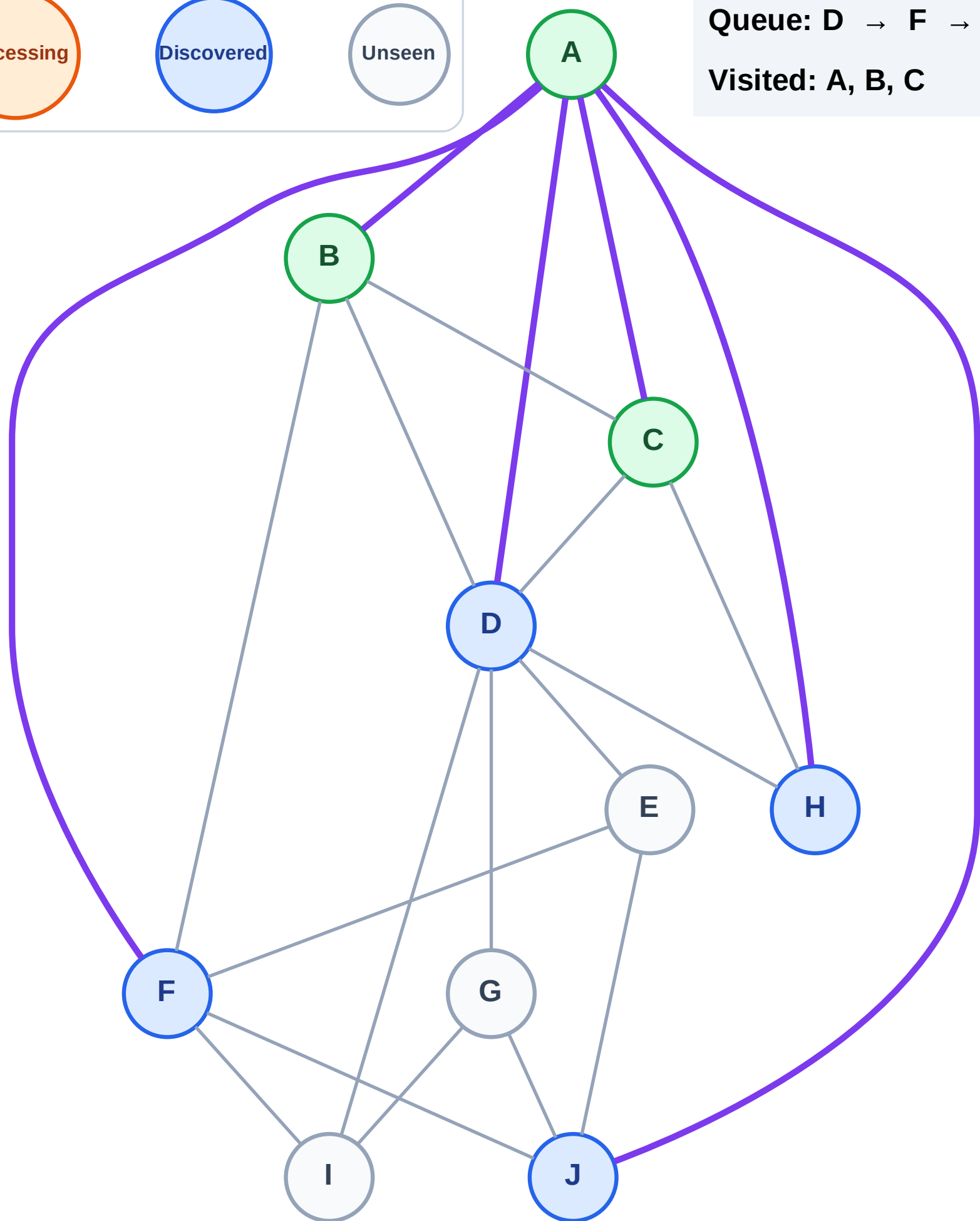
Step 7: No new neighbors from C

## Legend



Queue: D → F → H → J

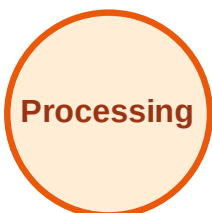
Visited: A, B, C



# BFS Traversal

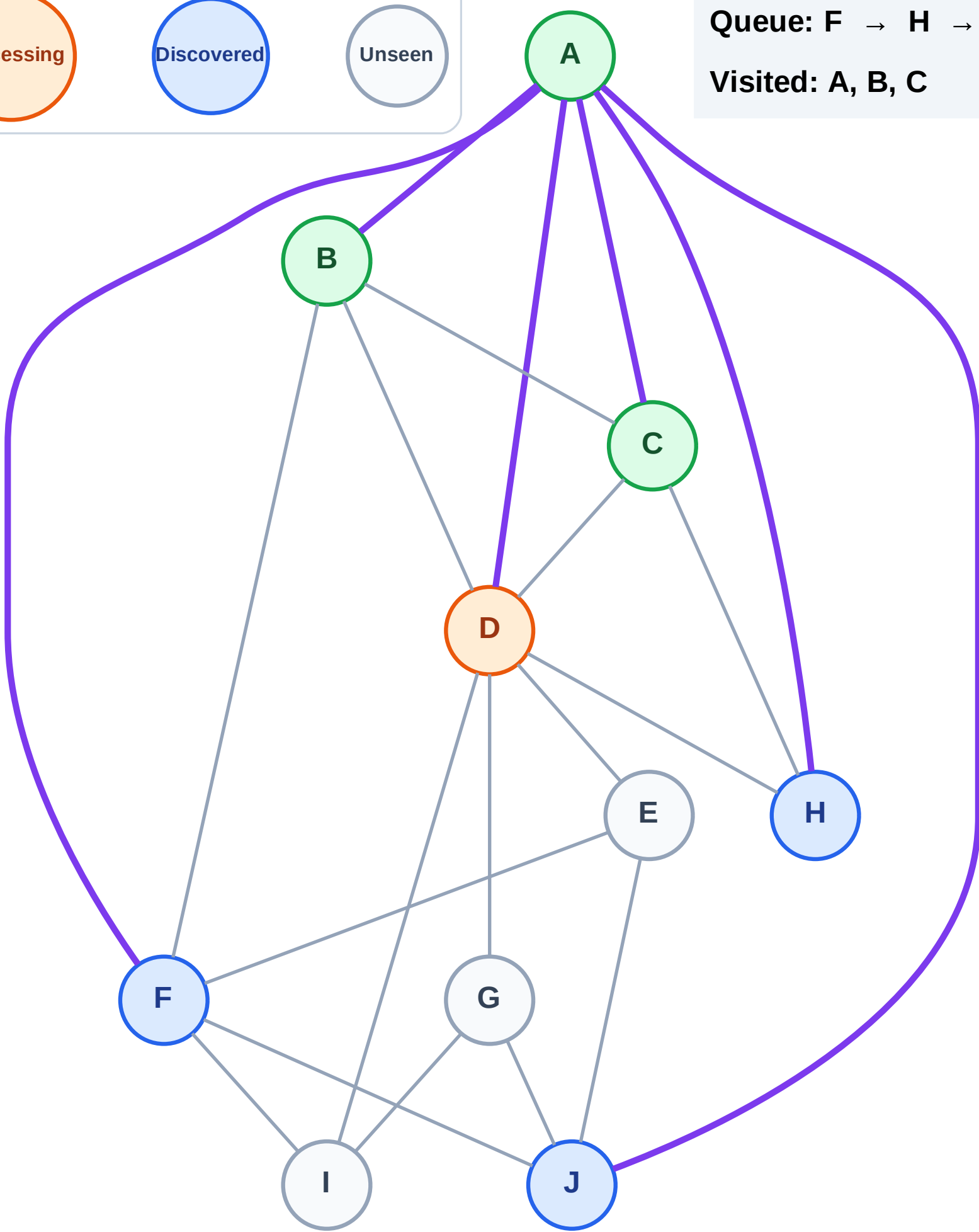
Step 8: Process node D

## Legend



Queue: F → H → J

Visited: A, B, C



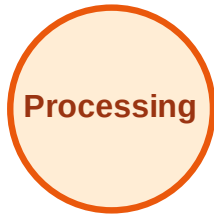
# BFS Traversal

Step 9: Discover E, G, I from D

## Legend



Complete



Processing

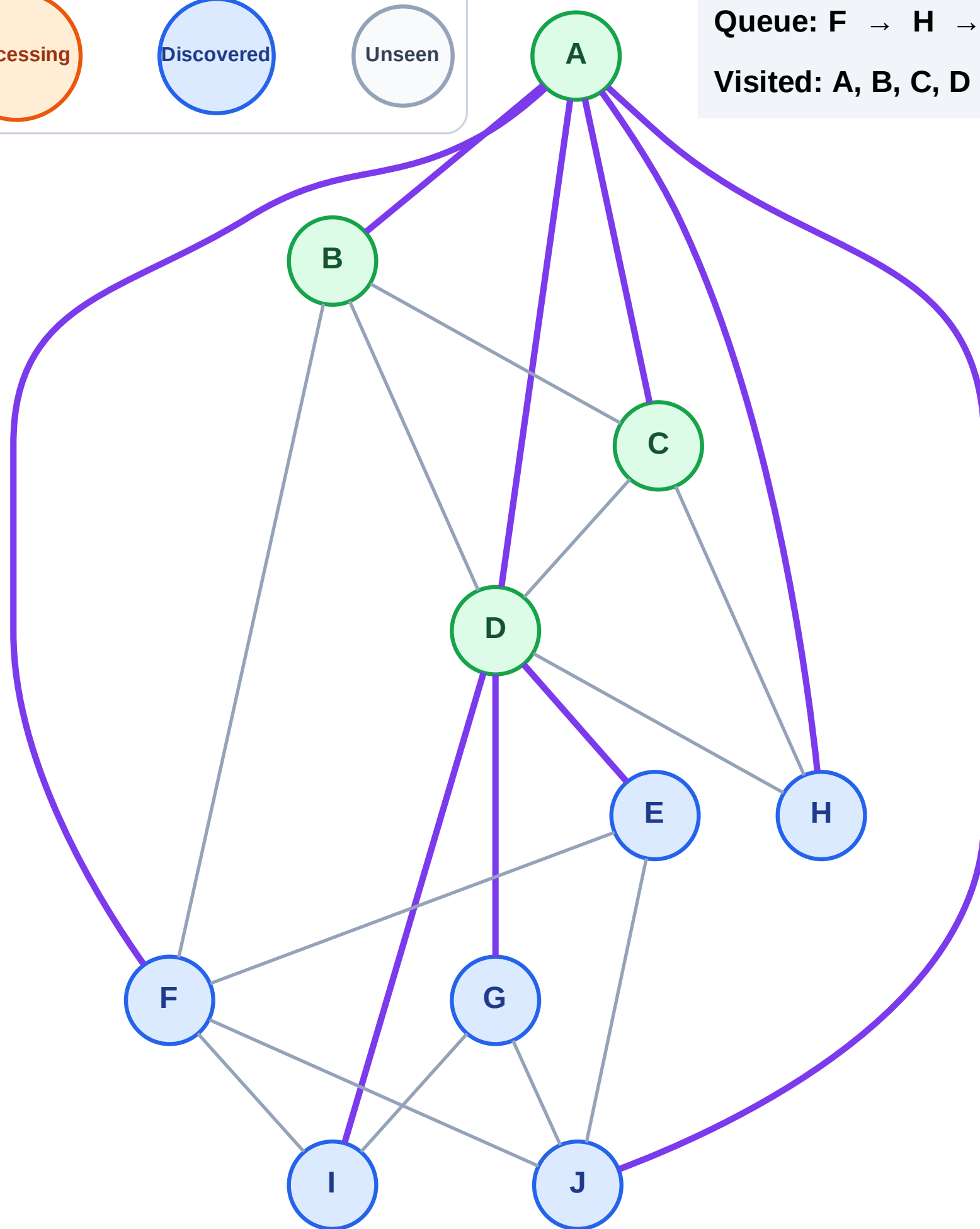


Discovered



Unseen

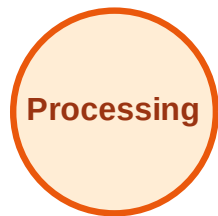
Queue: F → H → J → E → G → I  
Visited: A, B, C, D



# BFS Traversal

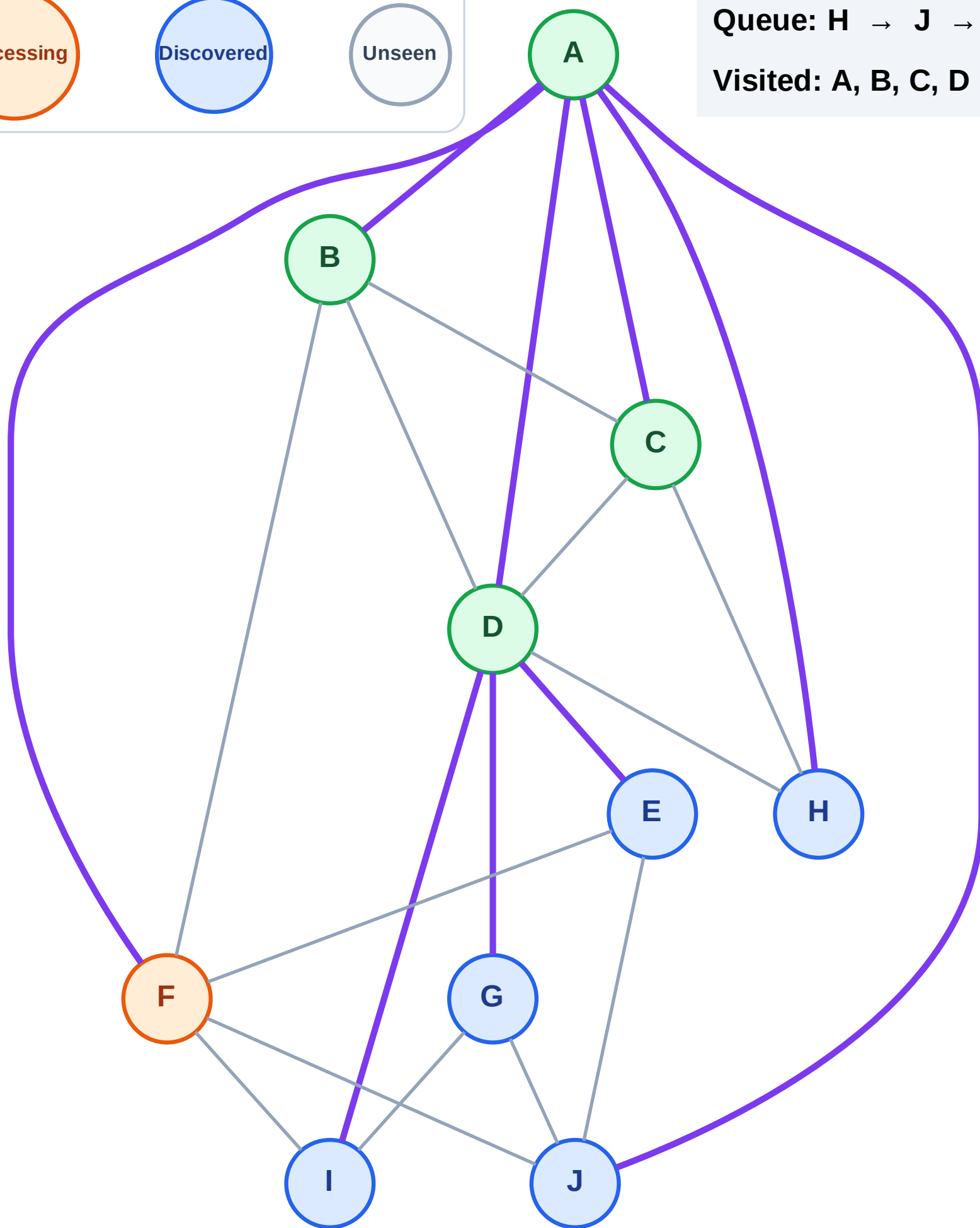
Step 10: Process node F

## Legend



Queue: H → J → E → G → I

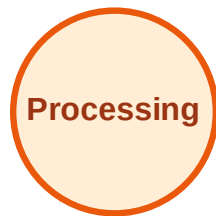
Visited: A, B, C, D



# BFS Traversal

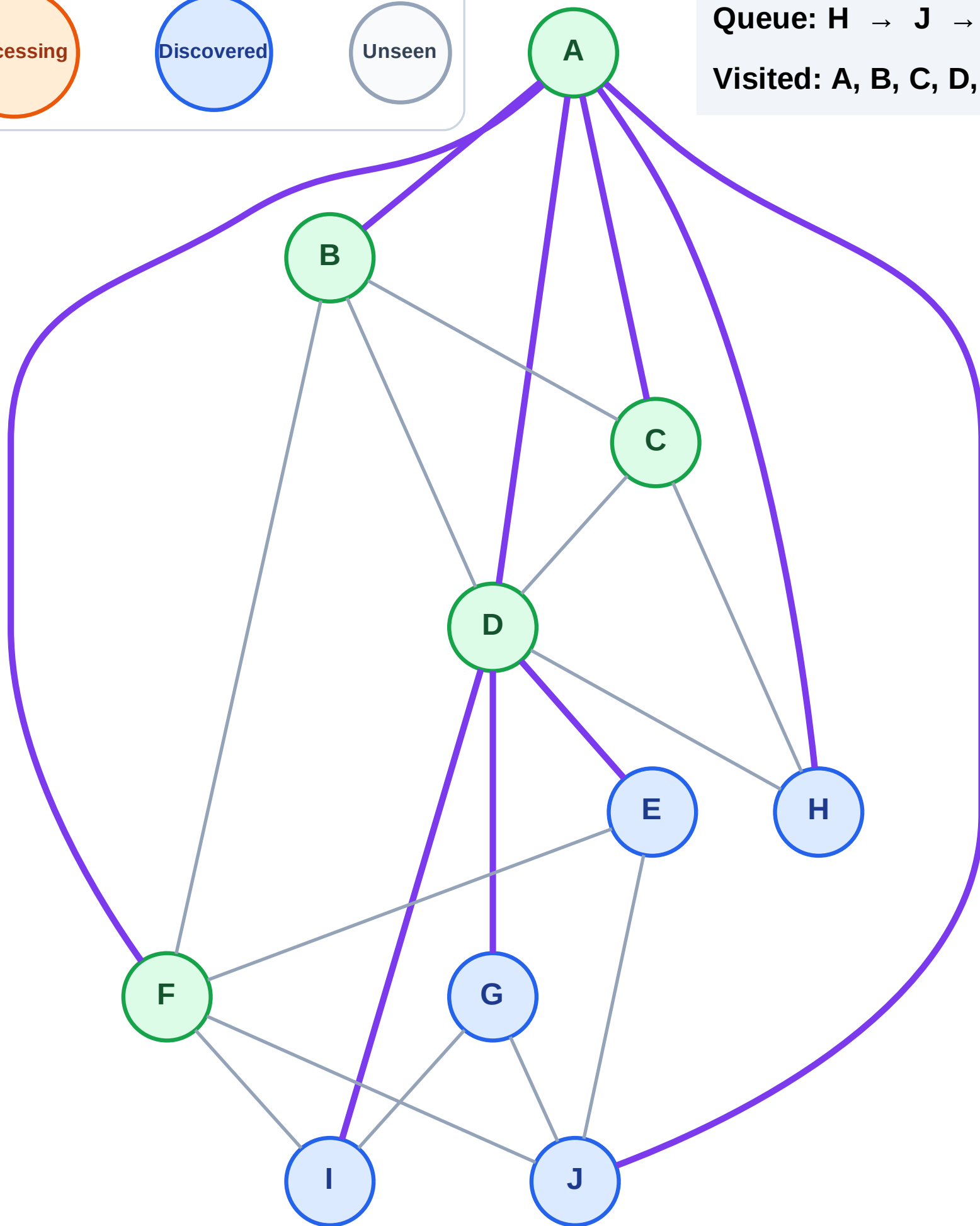
Step 11: No new neighbors from F

## Legend



Queue: H → J → E → G → I

Visited: A, B, C, D, F



# BFS Traversal

Step 12: Process node H

Legend

Complete

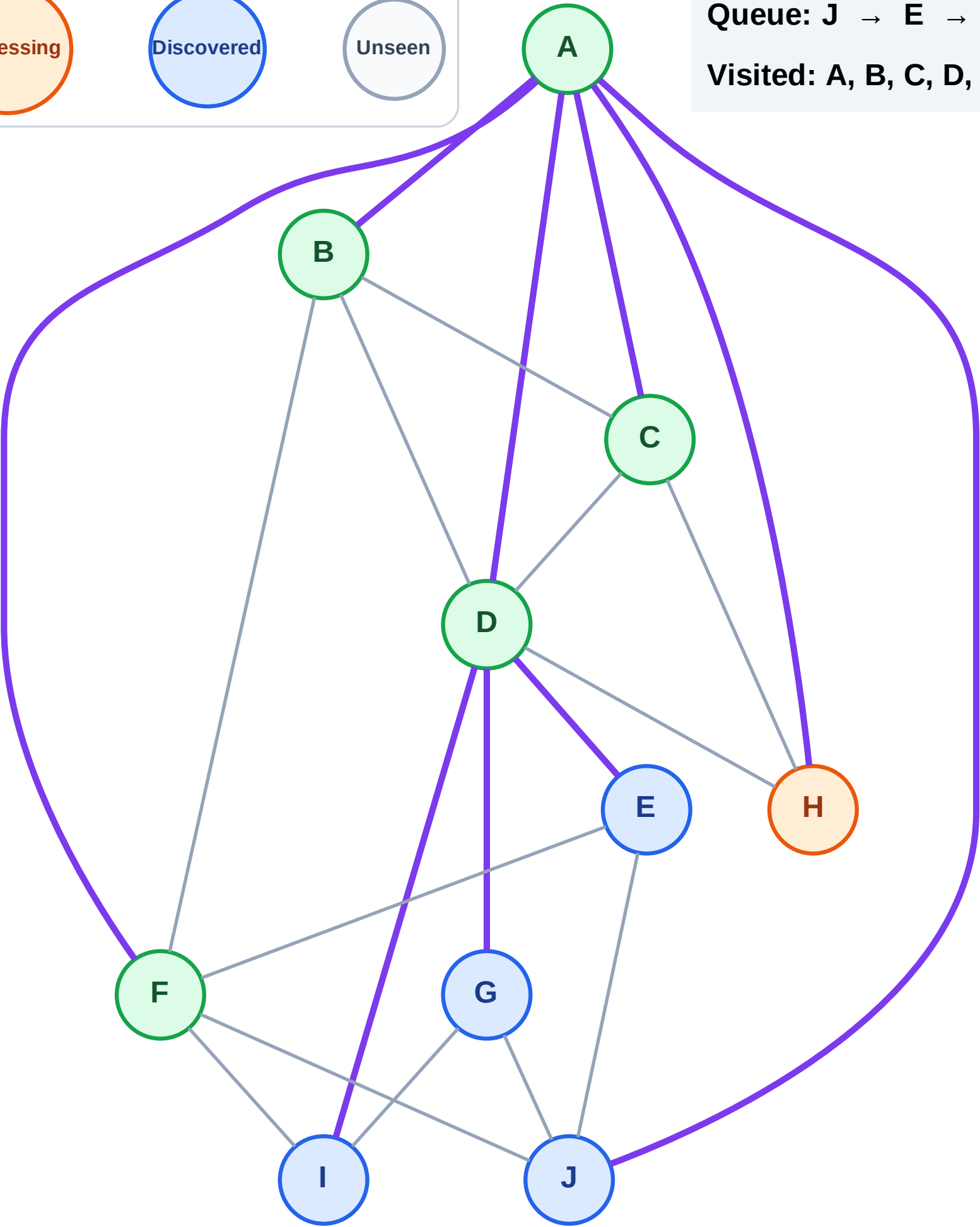
Processing

Discovered

Unseen

Queue: J → E → G → I

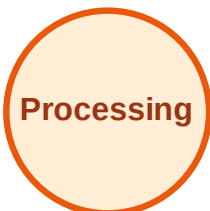
Visited: A, B, C, D, F



# BFS Traversal

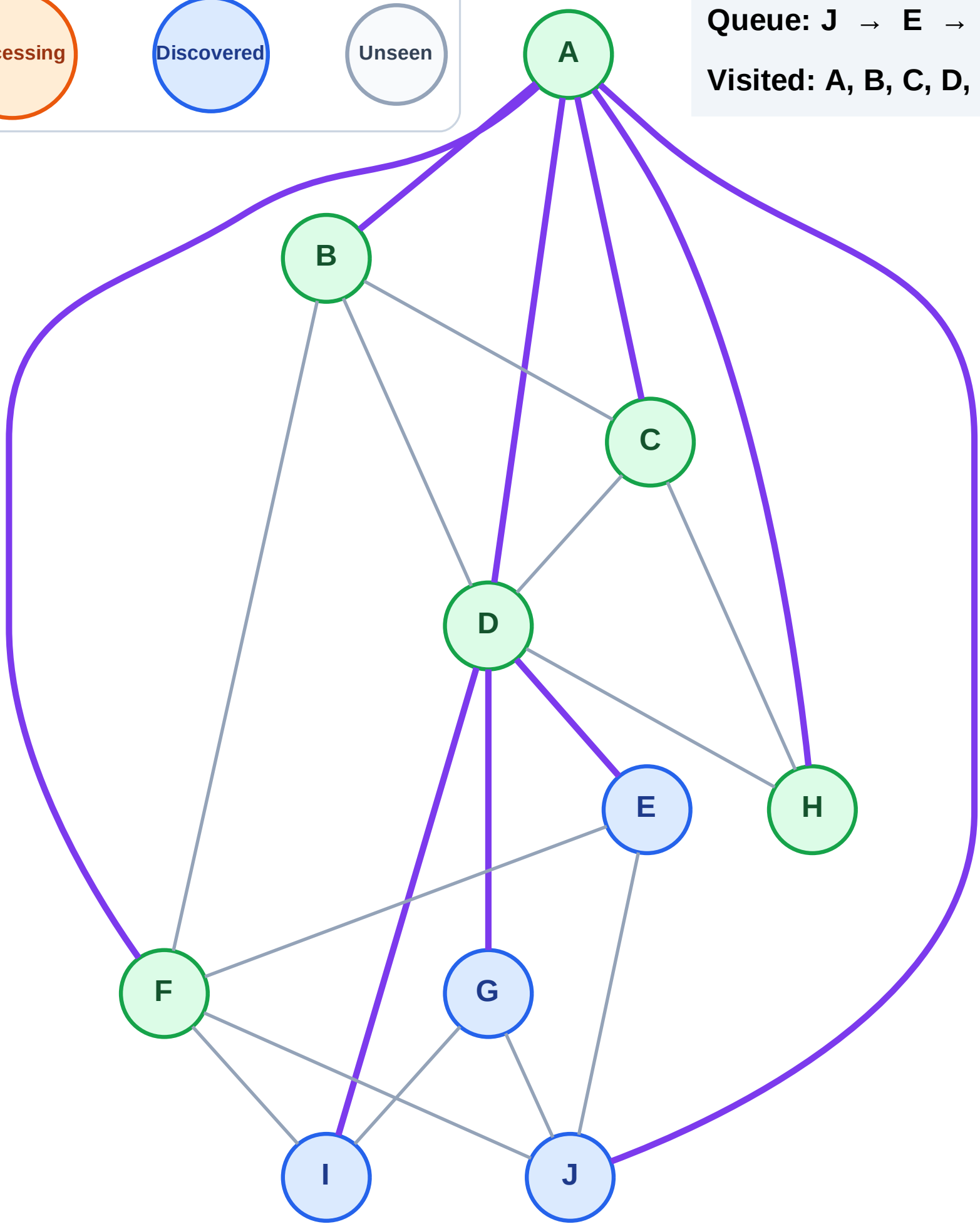
Step 13: No new neighbors from H

## Legend



Queue: J → E → G → I

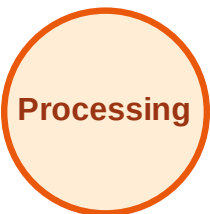
Visited: A, B, C, D, F, H



# BFS Traversal

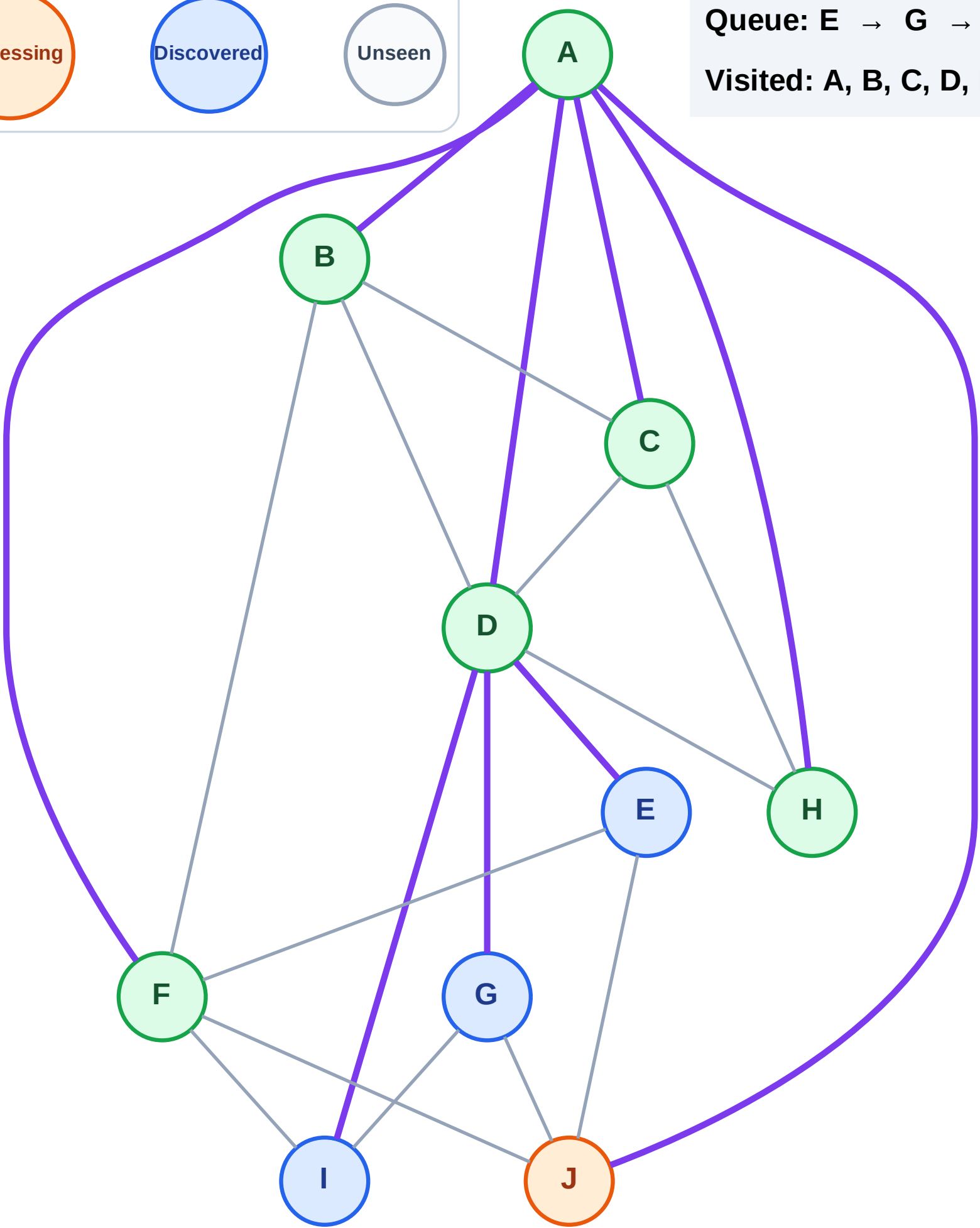
Step 14: Process node J

## Legend



Queue: E → G → I

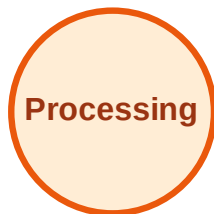
Visited: A, B, C, D, F, H



# BFS Traversal

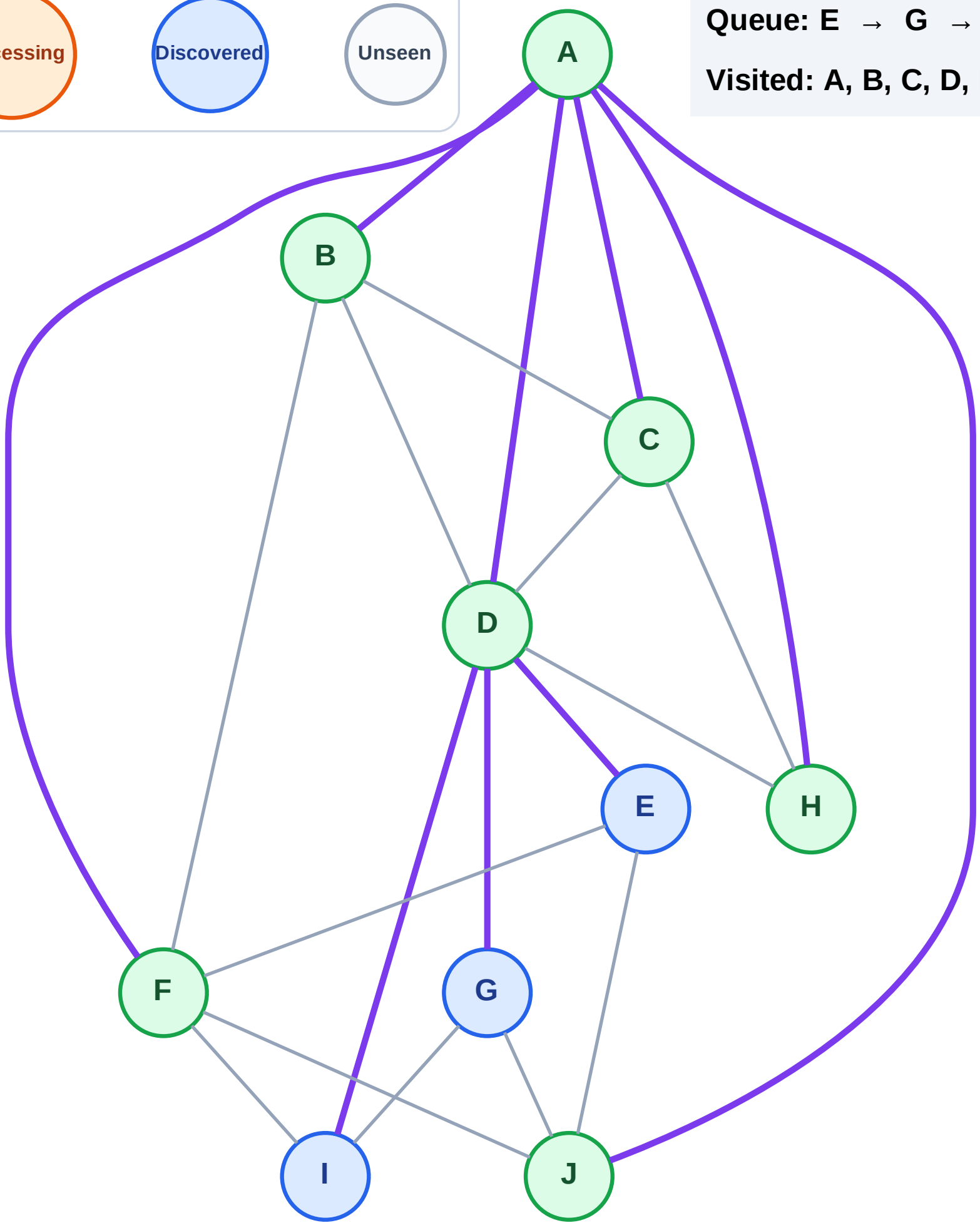
Step 15: No new neighbors from J

## Legend



Queue: E → G → I

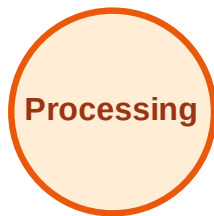
Visited: A, B, C, D, F, H, J



# BFS Traversal

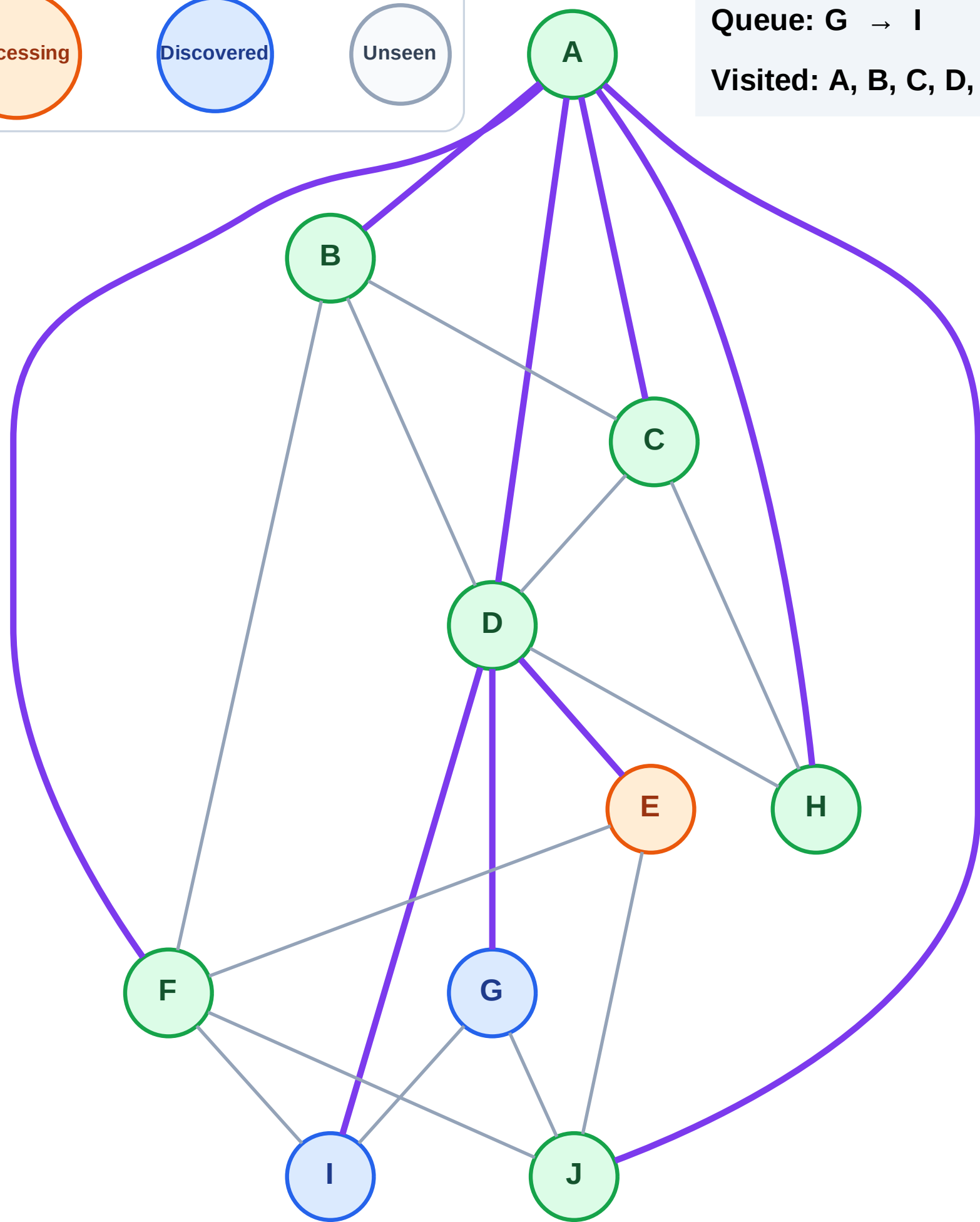
Step 16: Process node E

## Legend



Queue: G → I

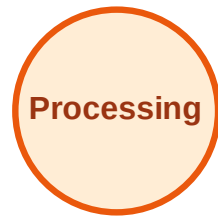
Visited: A, B, C, D, F, H, J



# BFS Traversal

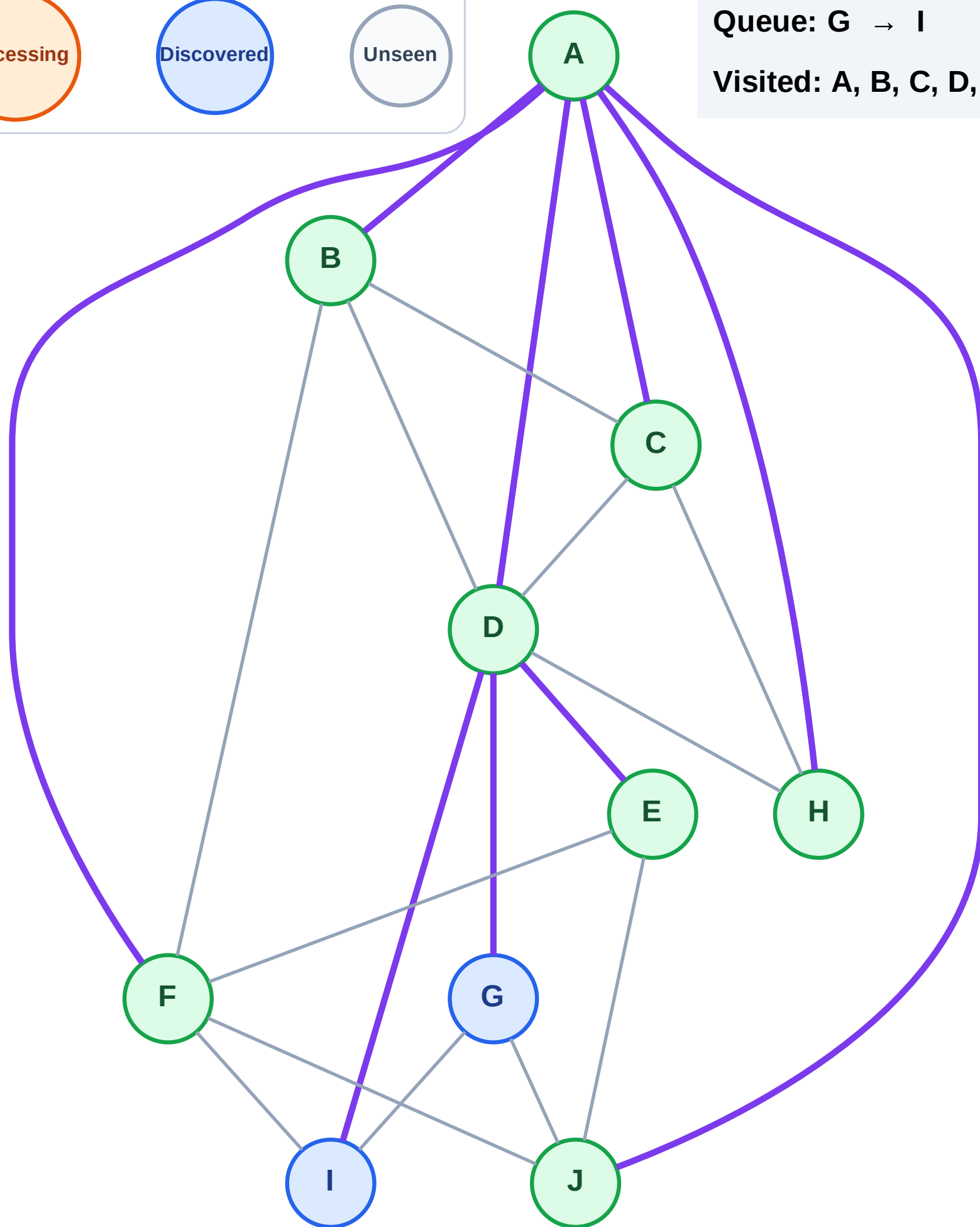
Step 17: No new neighbors from E

## Legend



Queue: G → I

Visited: A, B, C, D, E, F, H, J



# BFS Traversal

Step 18: Process node G

Legend

Complete

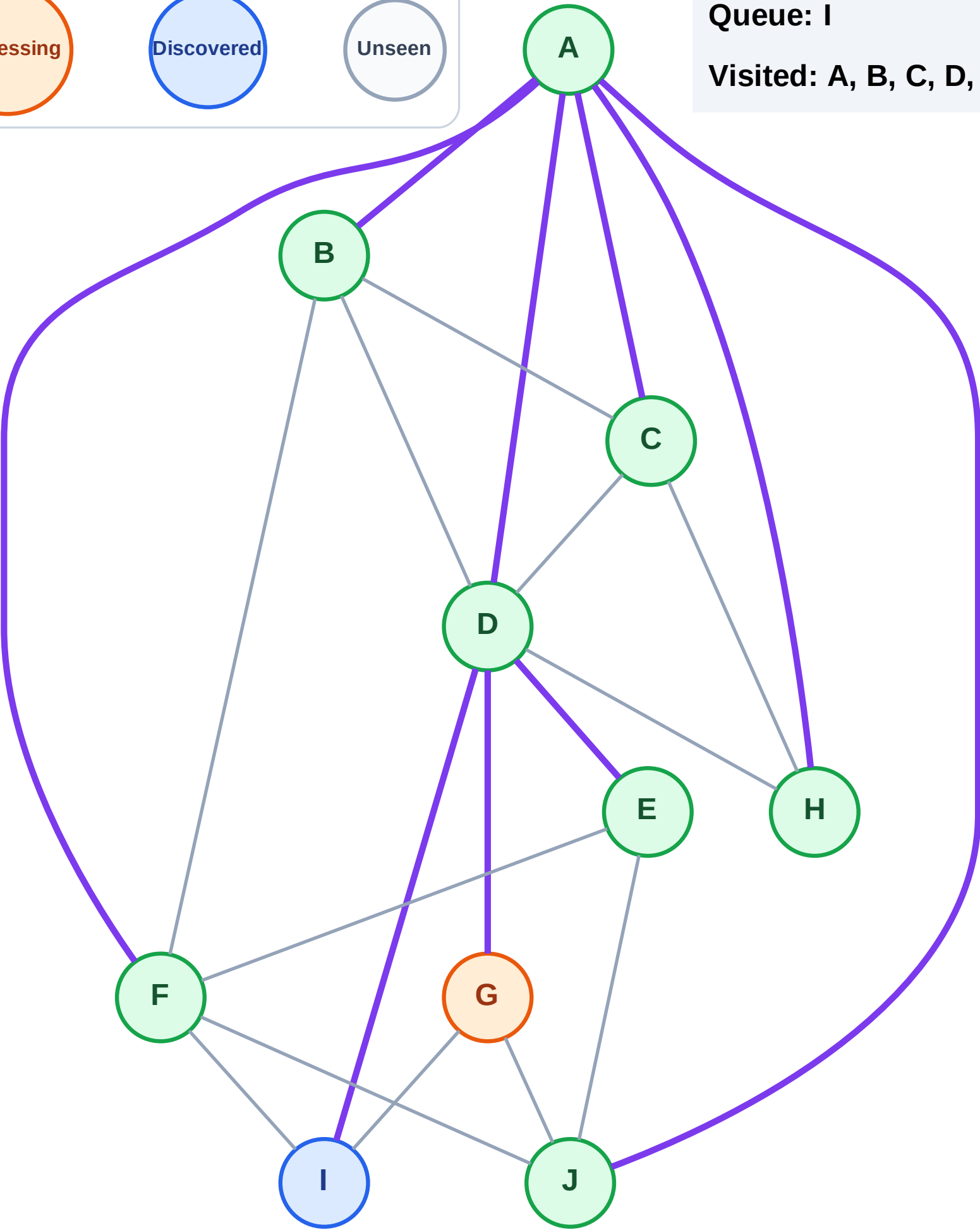
Processing

Discovered

Unseen

Queue: I

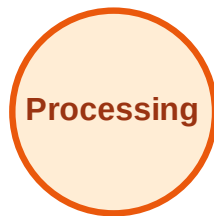
Visited: A, B, C, D, E, F, H, J



# BFS Traversal

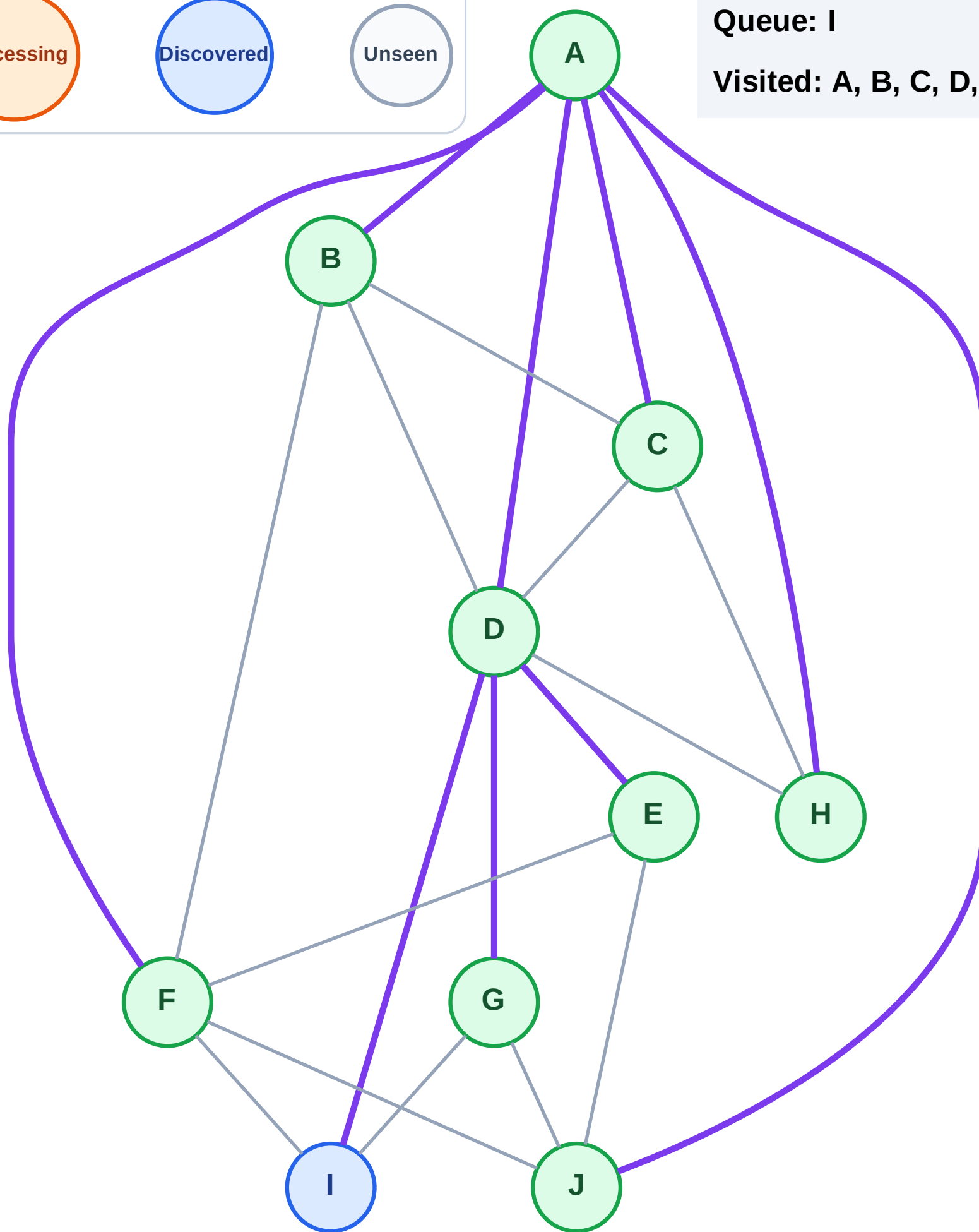
Step 19: No new neighbors from G

## Legend



Queue: I

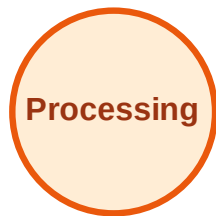
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

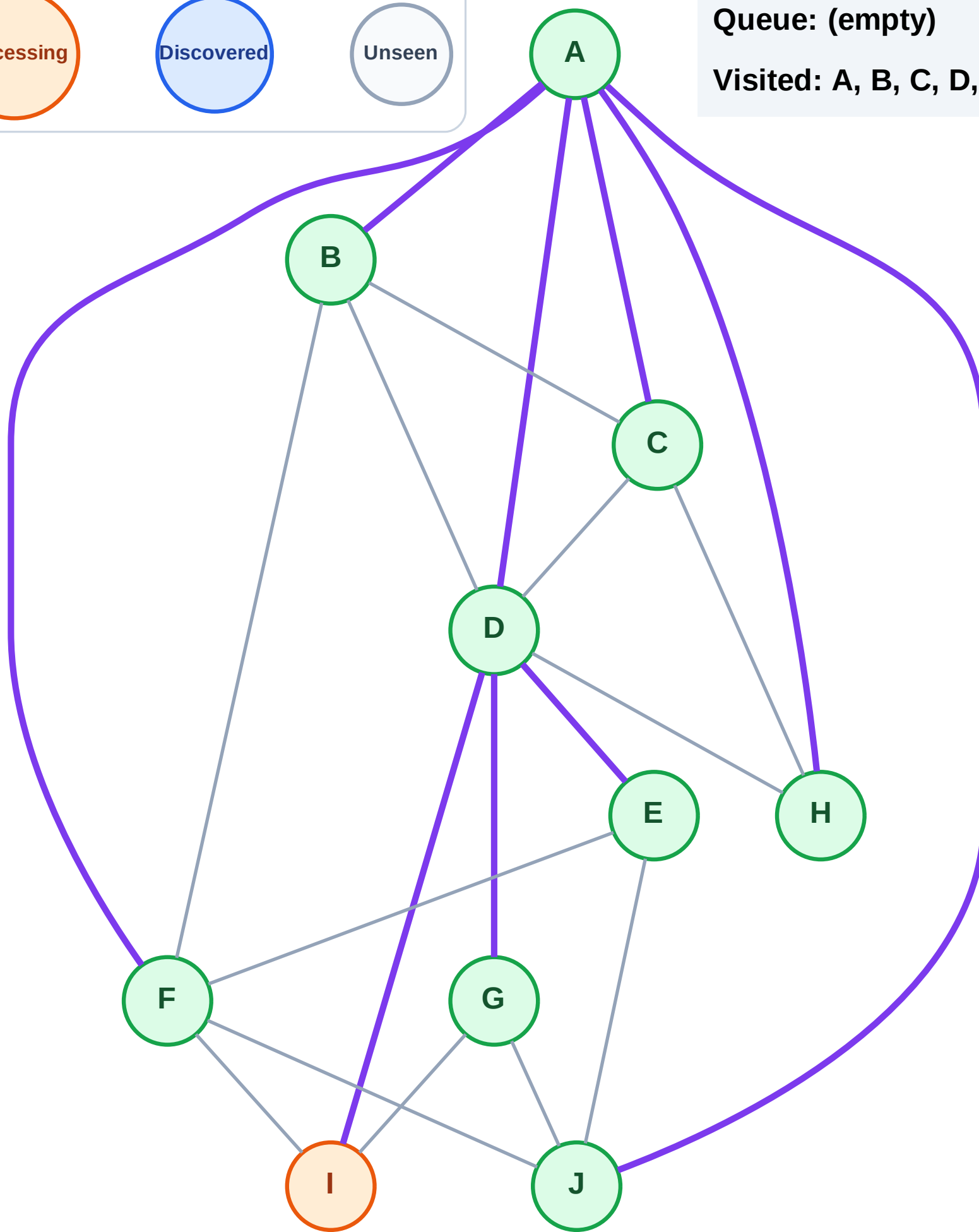
Step 20: Process node I

## Legend



Queue: (empty)

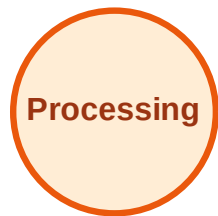
Visited: A, B, C, D, E, F, G, H, J



# BFS Traversal

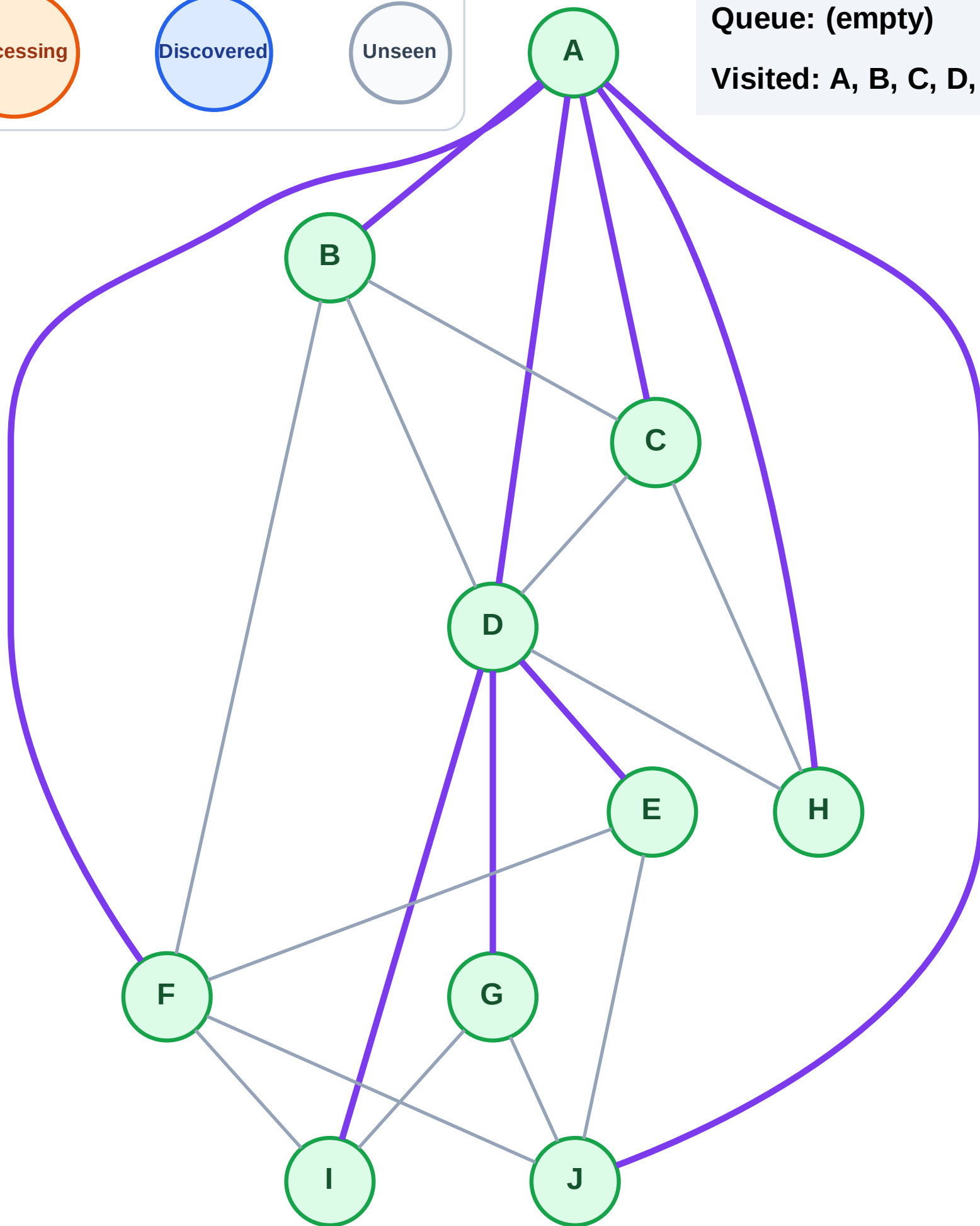
Step 21: No new neighbors from I

## Legend



Queue: (empty)

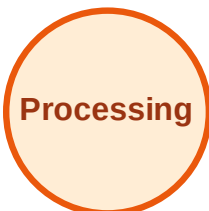
Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

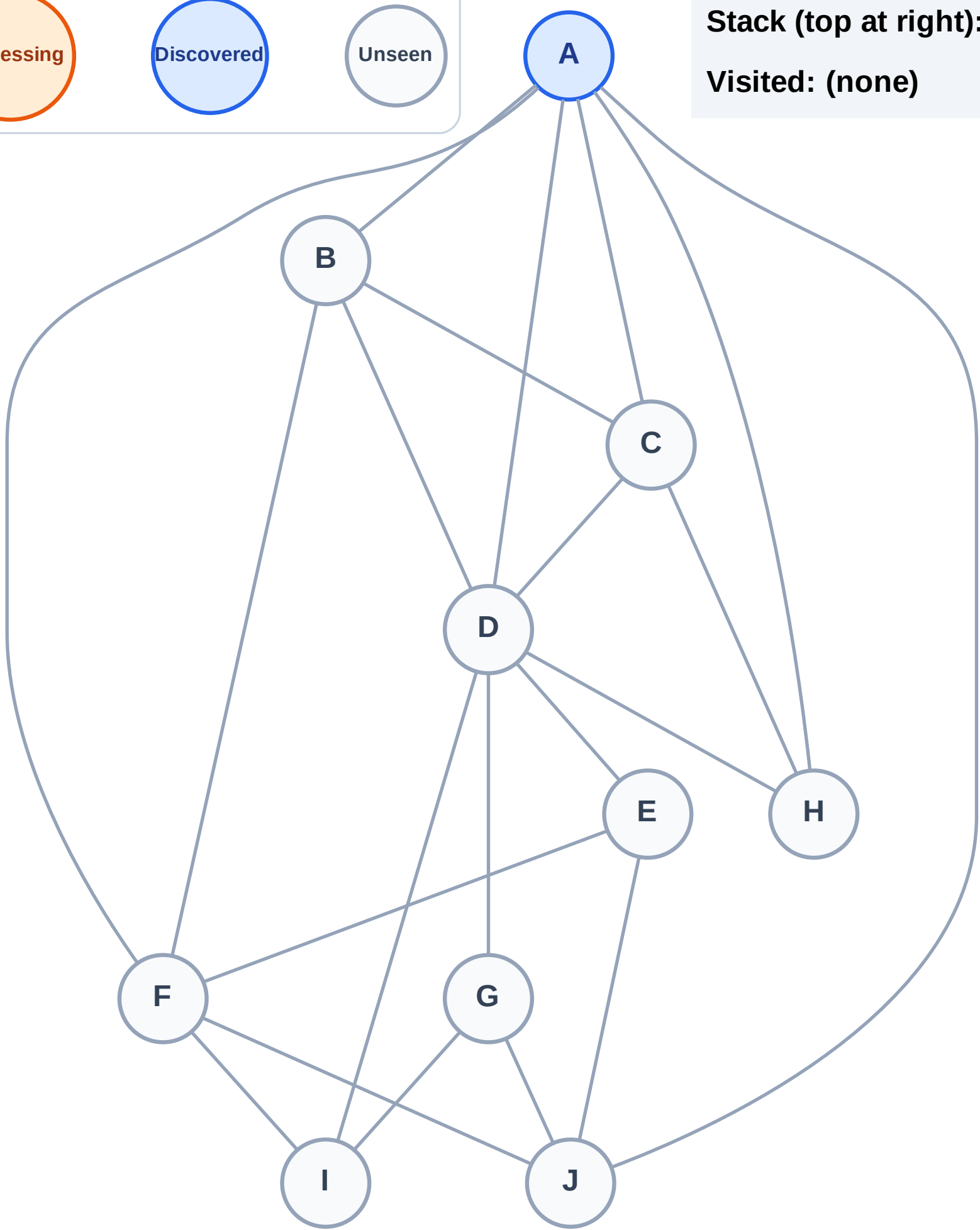
Step 1: Start at A

## Legend



Stack (top at right): A

Visited: (none)



# DFS Traversal

Step 2: Process node A

Legend

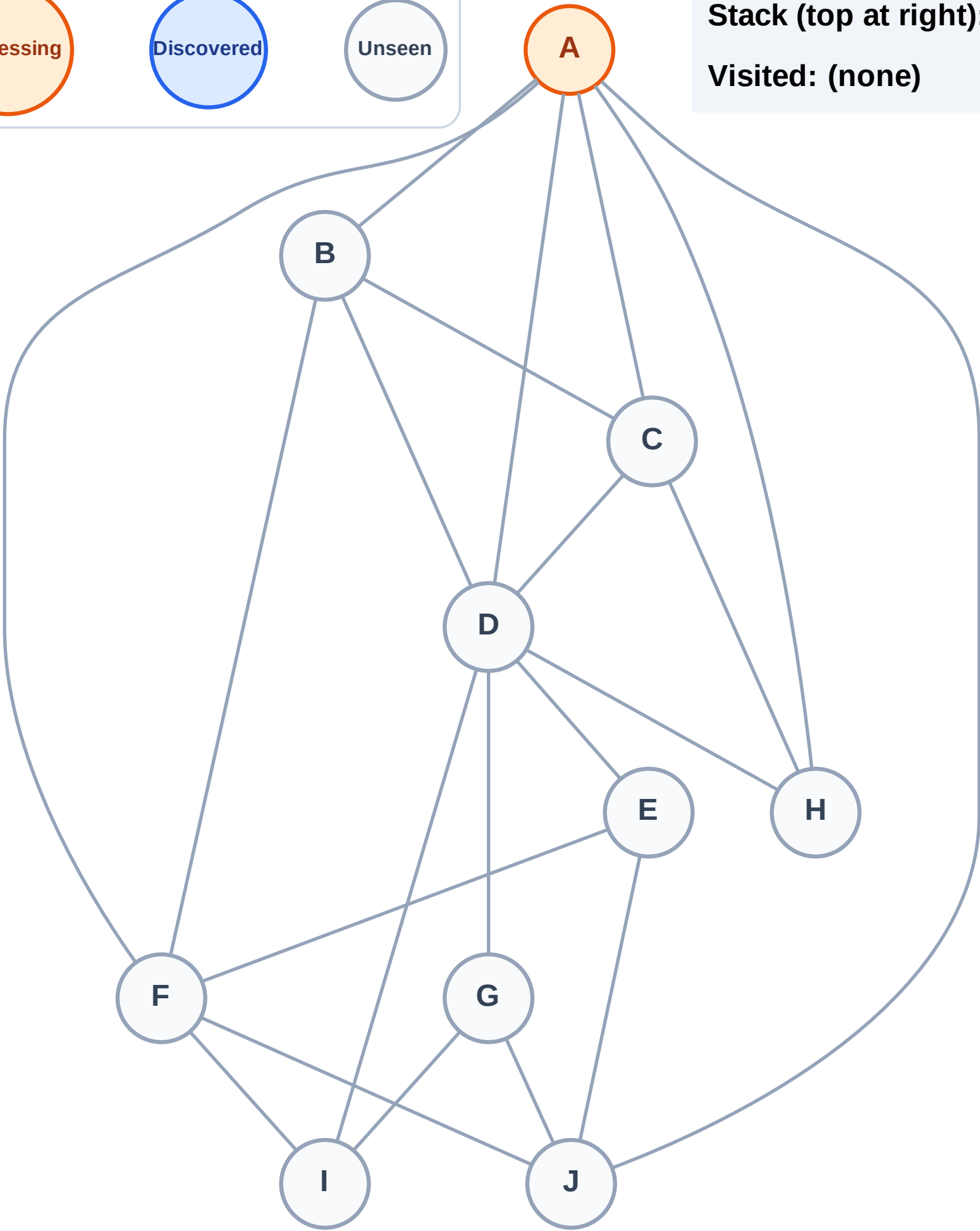
Complete

Processing

Discovered

Unseen

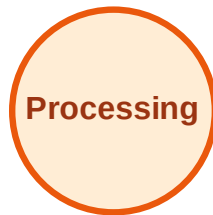
Stack (top at right): (empty)  
Visited: (none)



# DFS Traversal

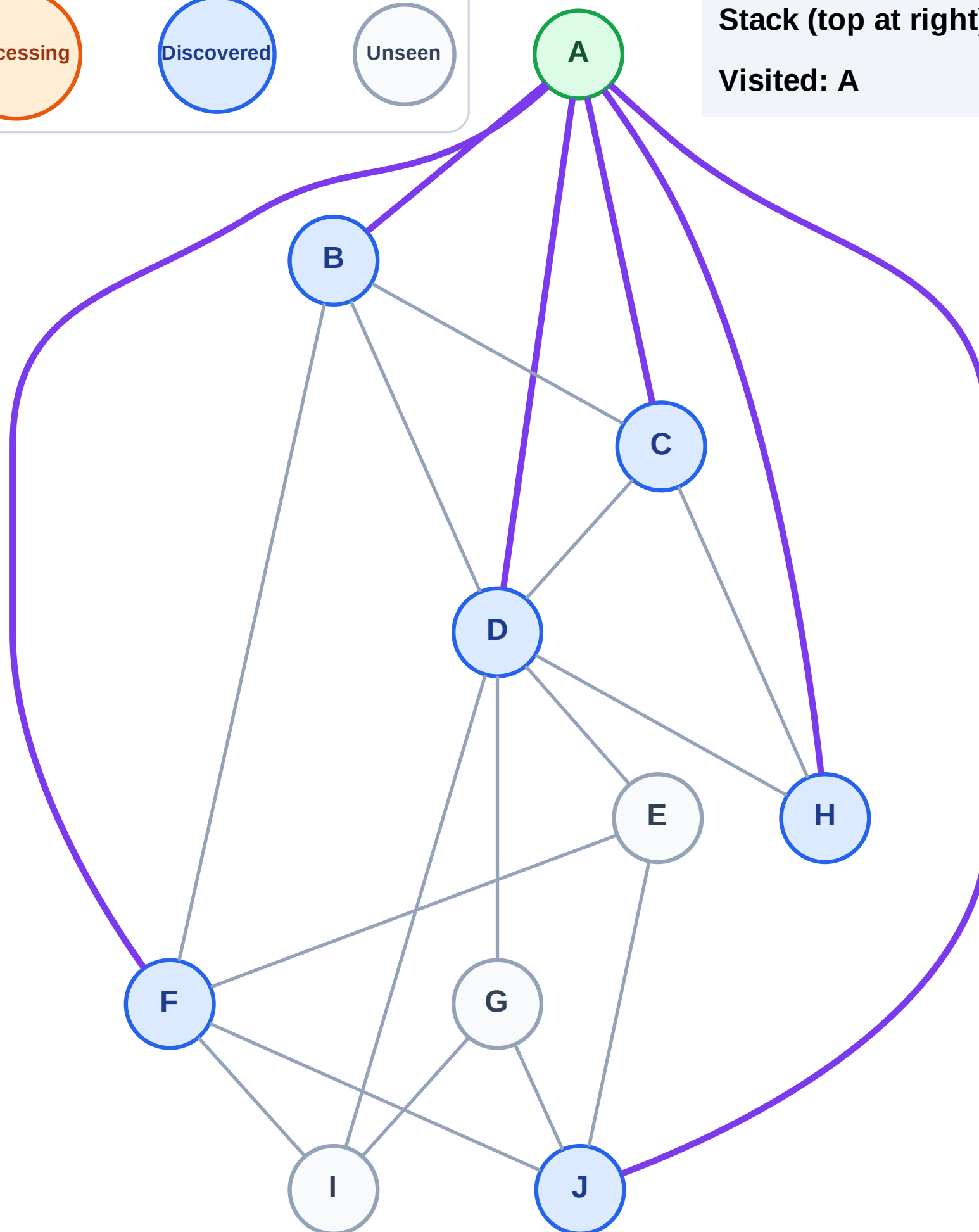
Step 3: Discover J, H, F, D, C, B from A

## Legend



Stack (top at right): J | H | F | D | C | B

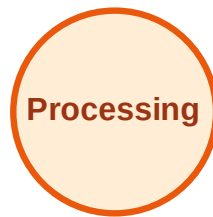
Visited: A



# DFS Traversal

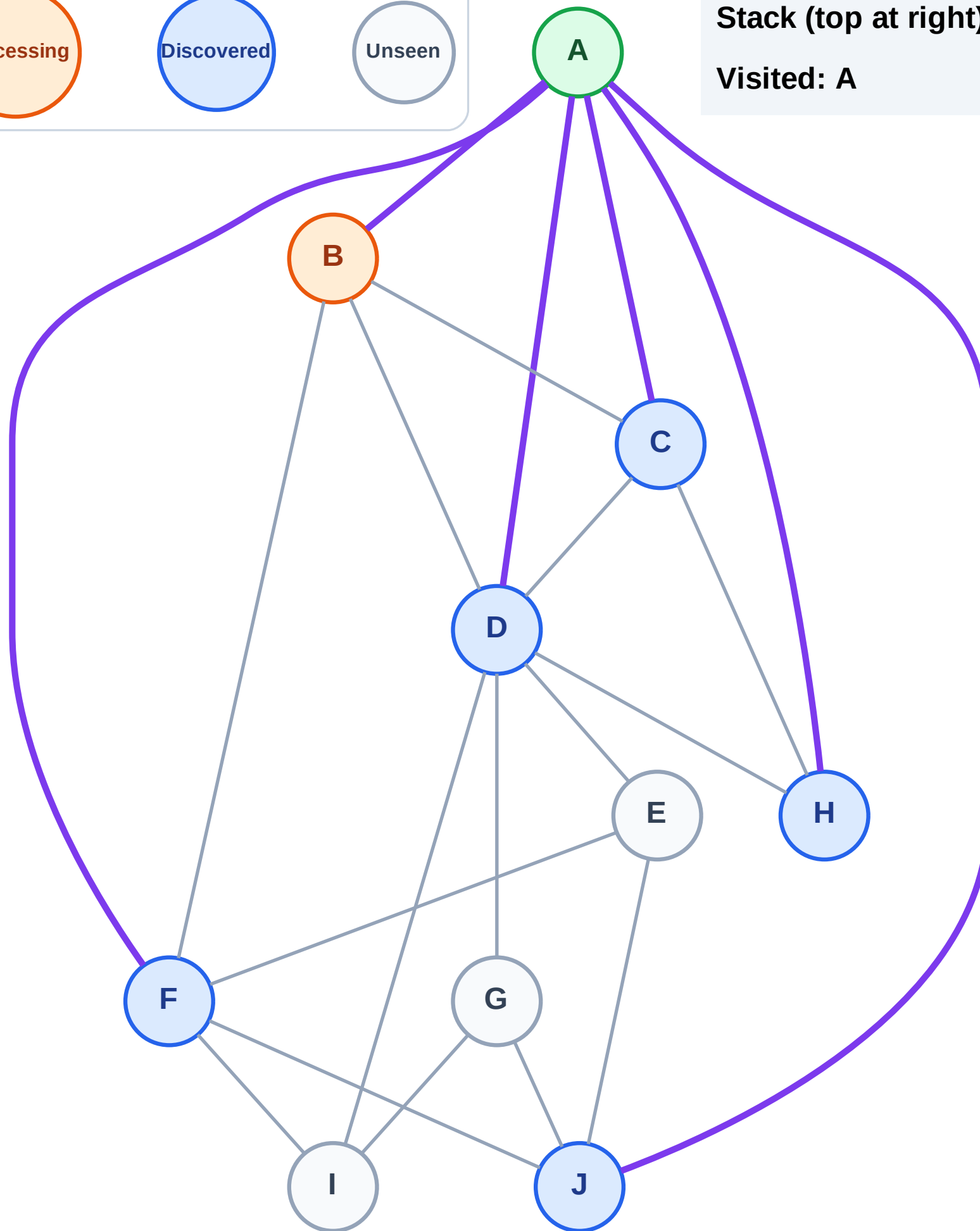
Step 4: Process node B

## Legend



Stack (top at right): J | H | F | D | C

Visited: A



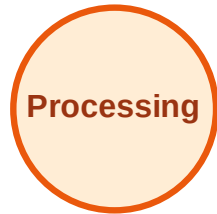
# DFS Traversal

Step 5: No new neighbors from B

## Legend



Complete



Processing



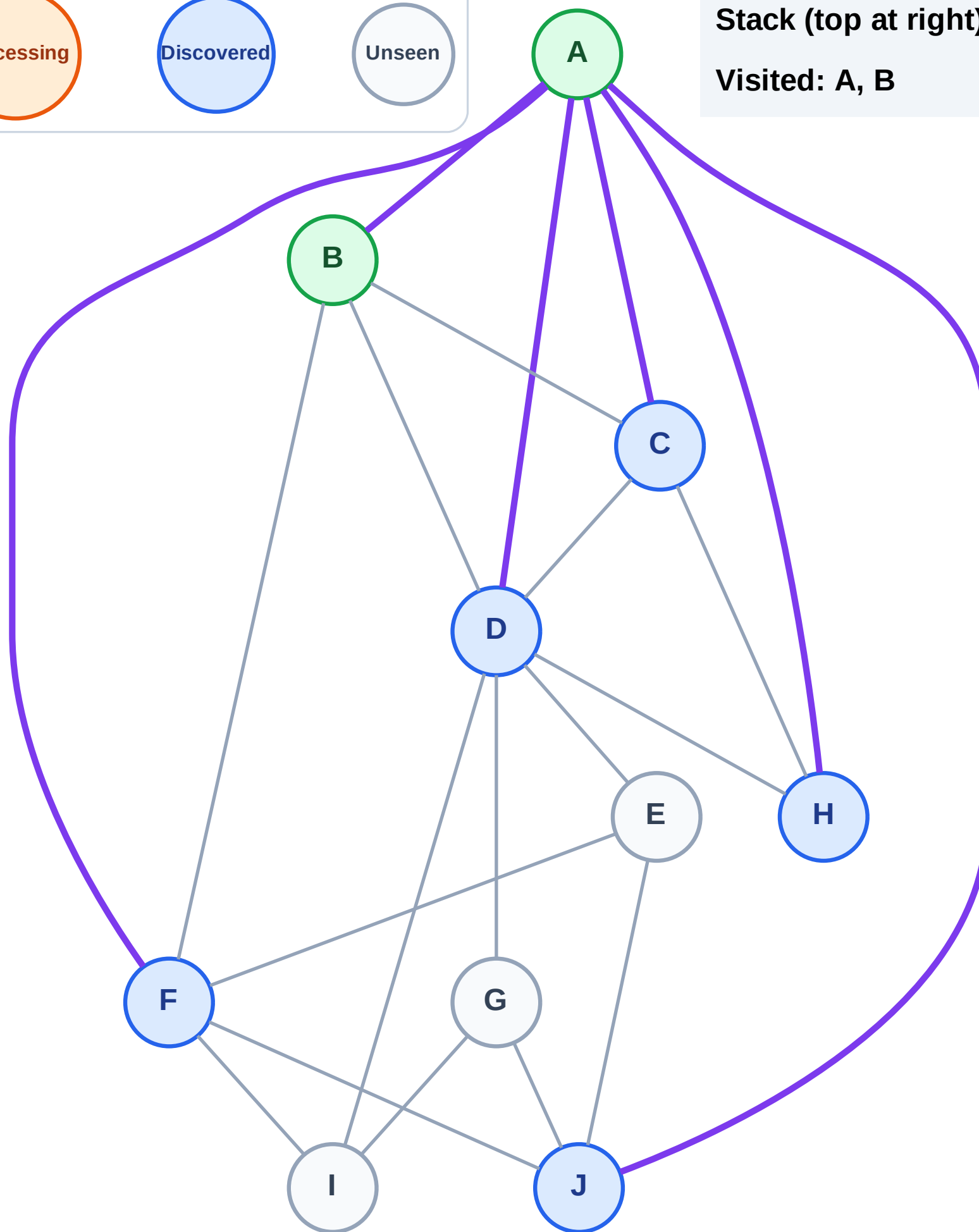
Discovered



Unseen

Stack (top at right): J | H | F | D | C

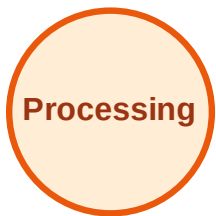
Visited: A, B



# DFS Traversal

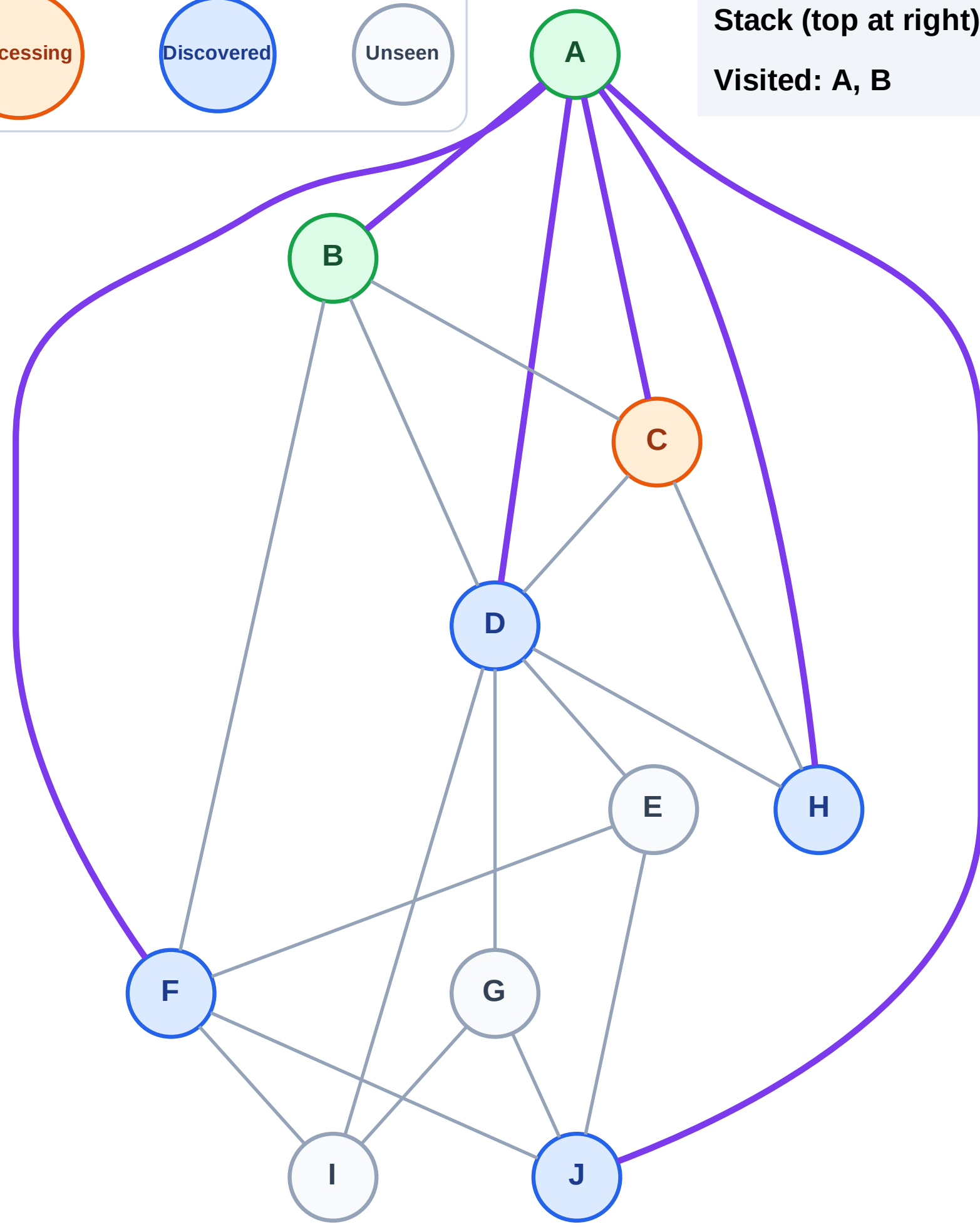
Step 6: Process node C

## Legend



Stack (top at right): J | H | F | D

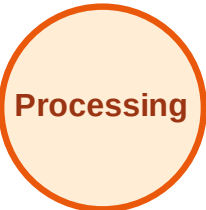
Visited: A, B



# DFS Traversal

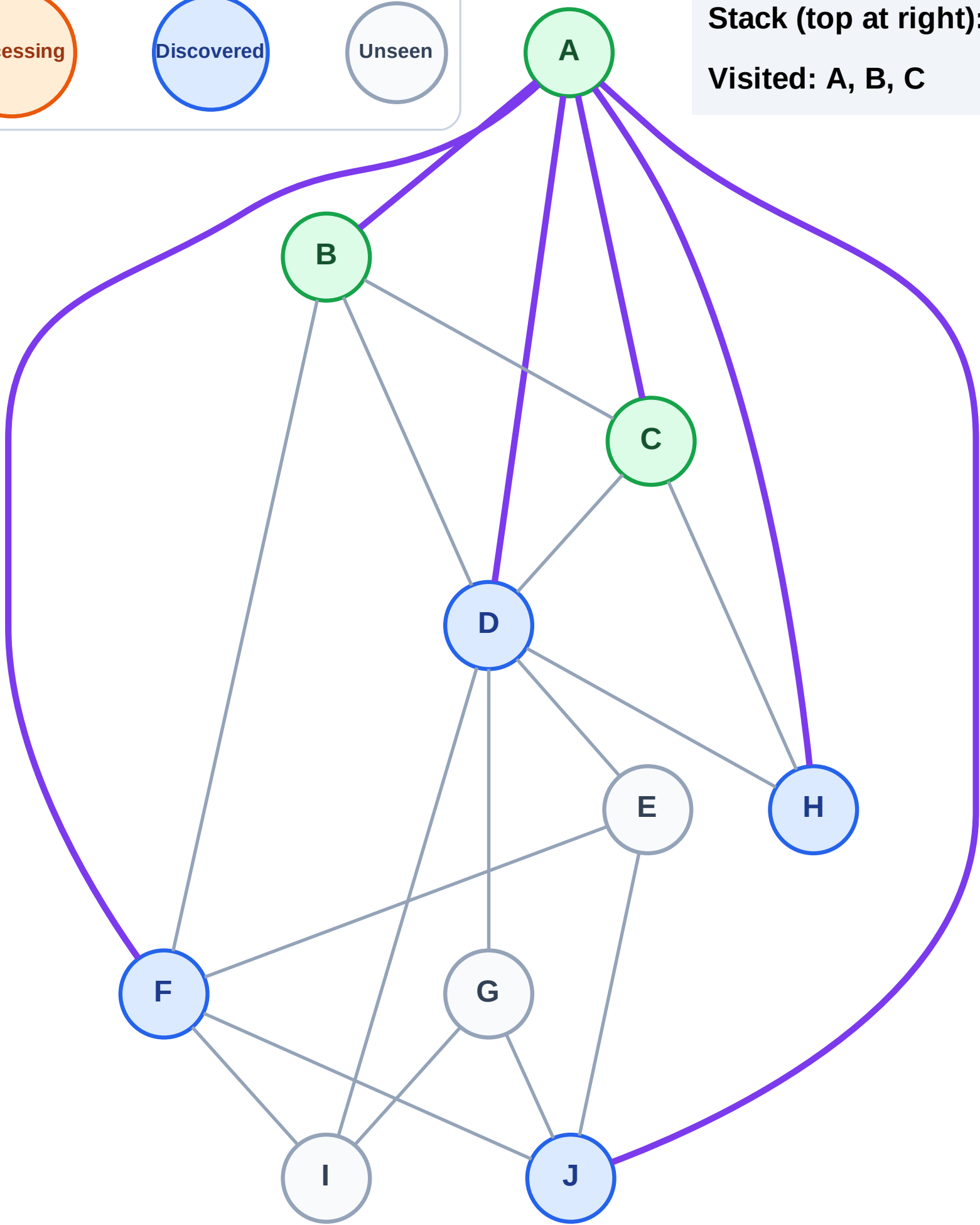
Step 7: No new neighbors from C

## Legend



Stack (top at right): J | H | F | D

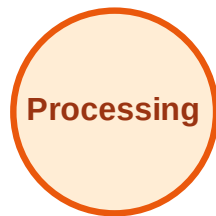
Visited: A, B, C



# DFS Traversal

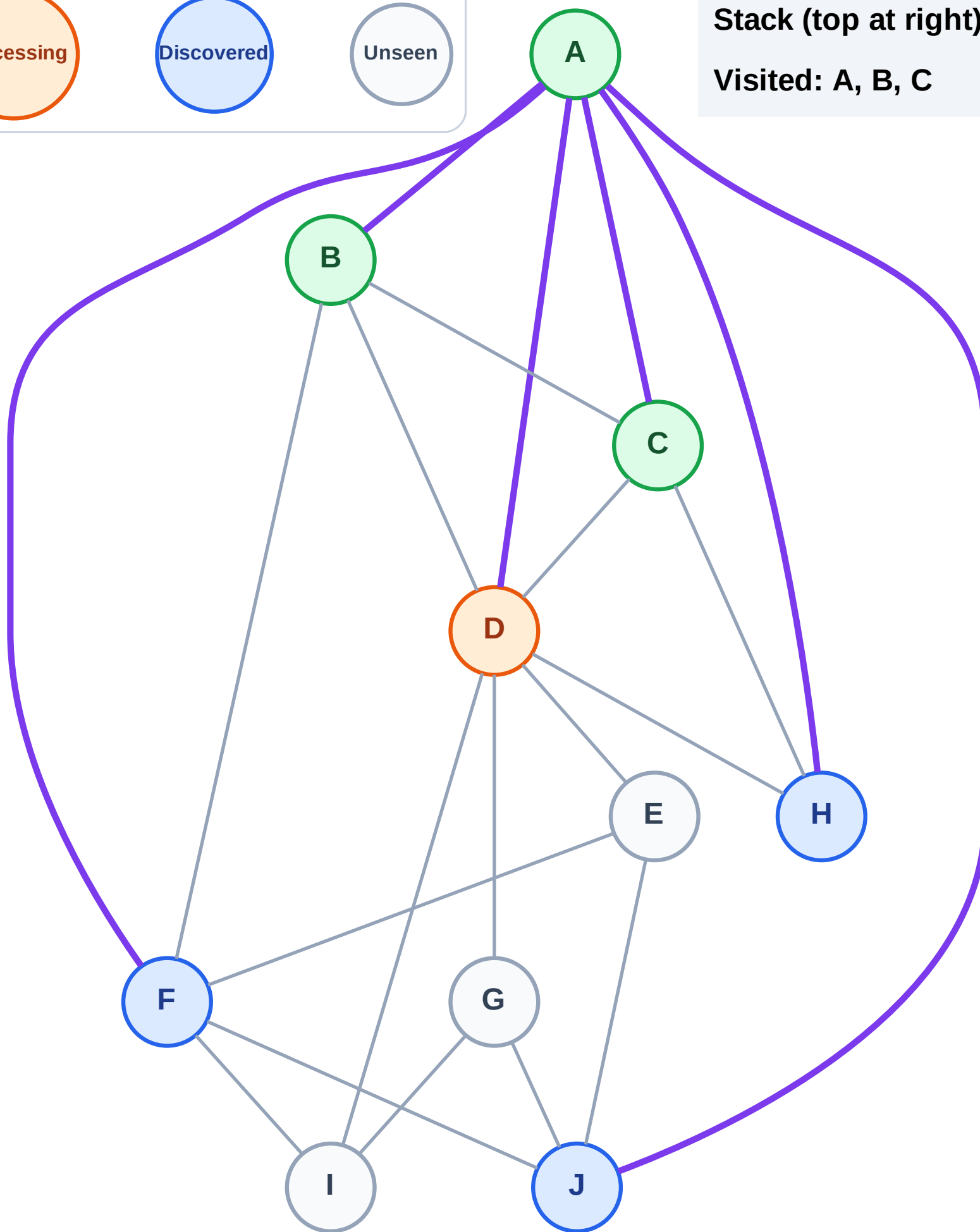
Step 8: Process node D

## Legend



Stack (top at right): J | H | F

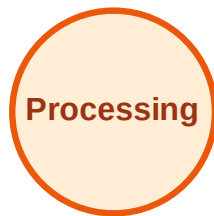
Visited: A, B, C



# DFS Traversal

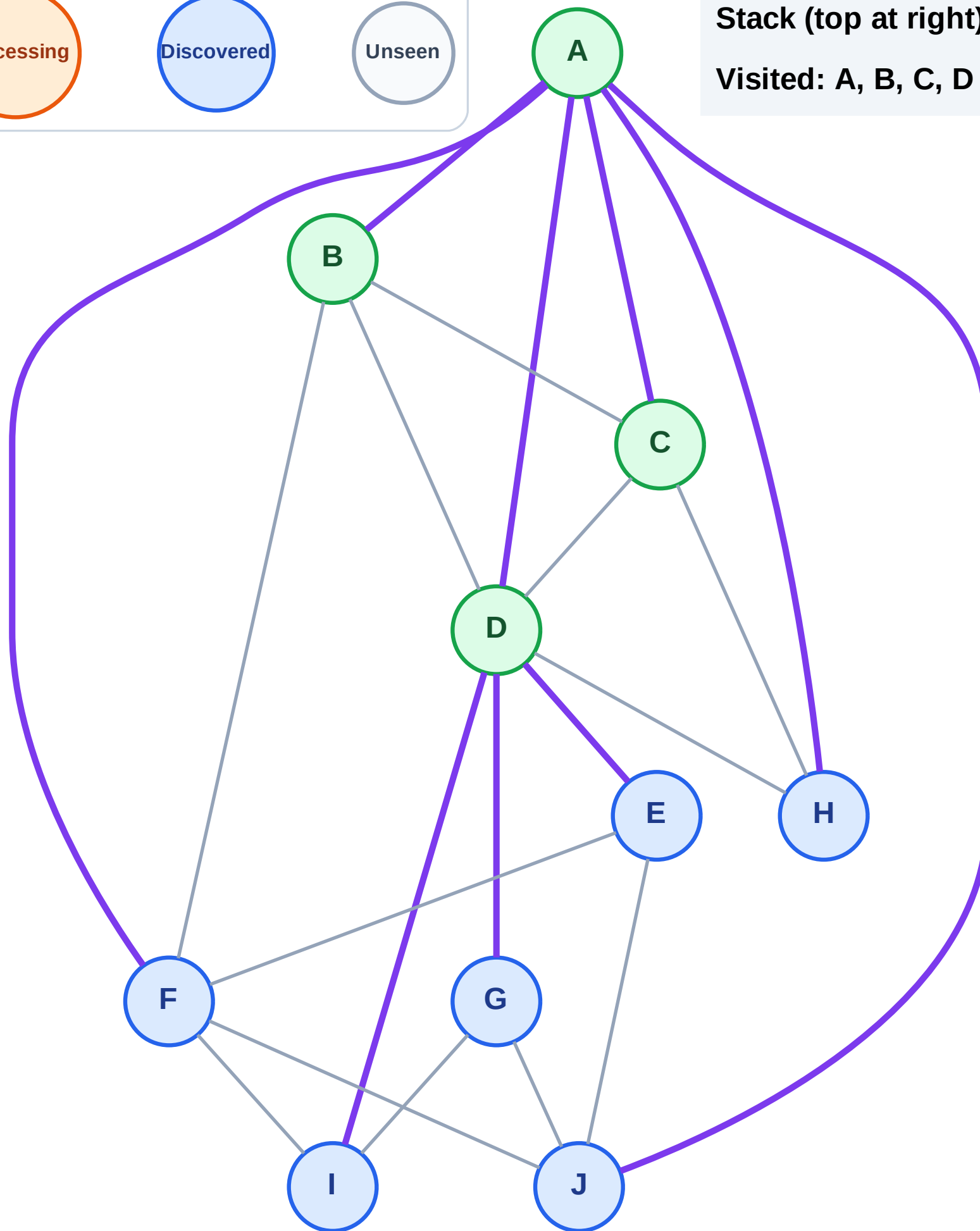
Step 9: Discover I, G, E from D

## Legend



Stack (top at right): J | H | F | I | G | E

Visited: A, B, C, D



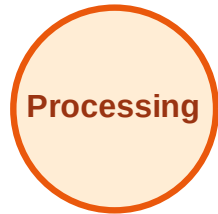
# DFS Traversal

Step 10: Process node E

## Legend



Complete



Processing



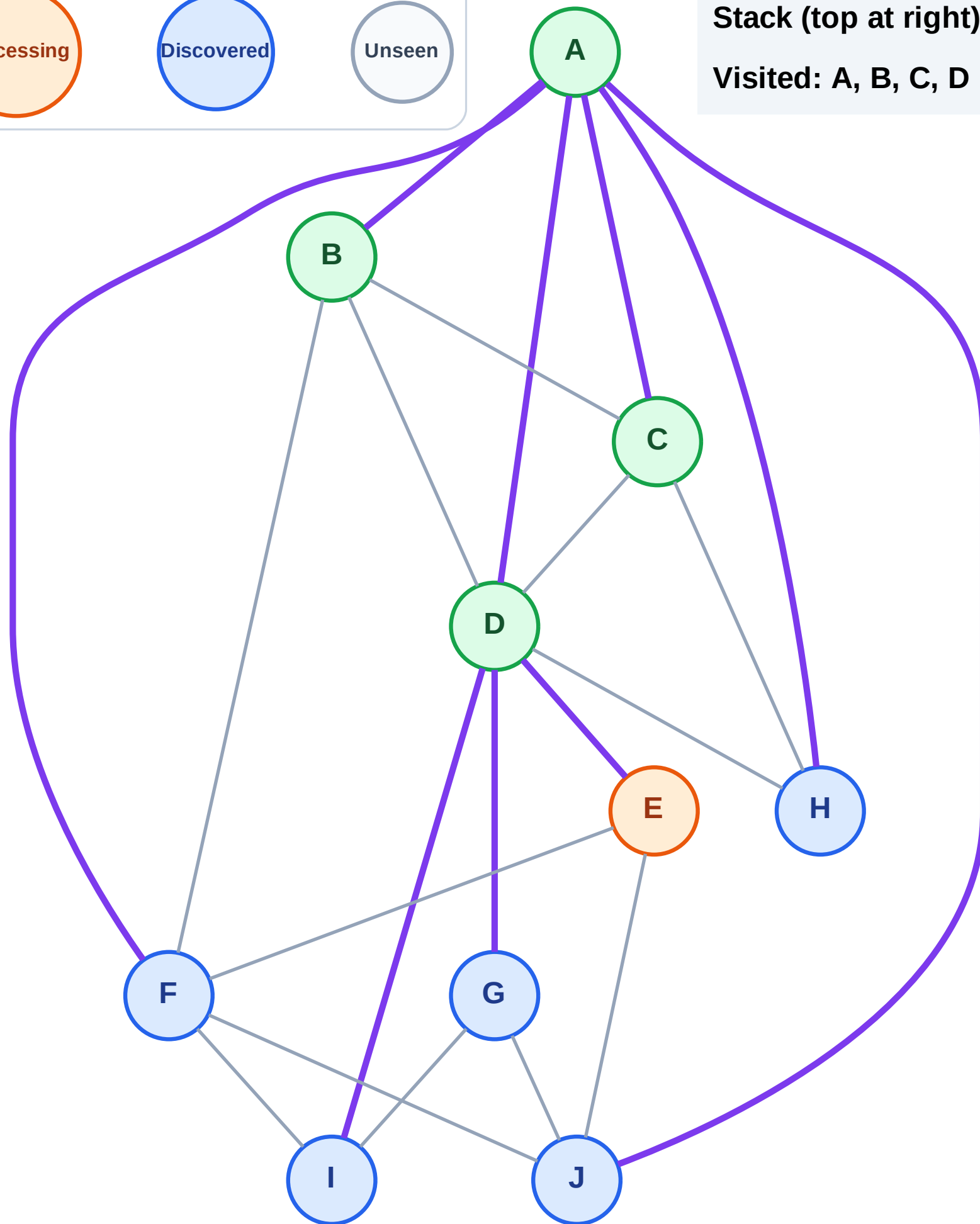
Discovered



Unseen

Stack (top at right): J | H | F | I | G

Visited: A, B, C, D



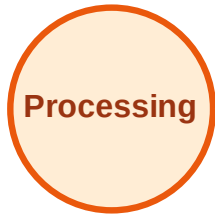
# DFS Traversal

Step 11: No new neighbors from E

## Legend



Complete



Processing



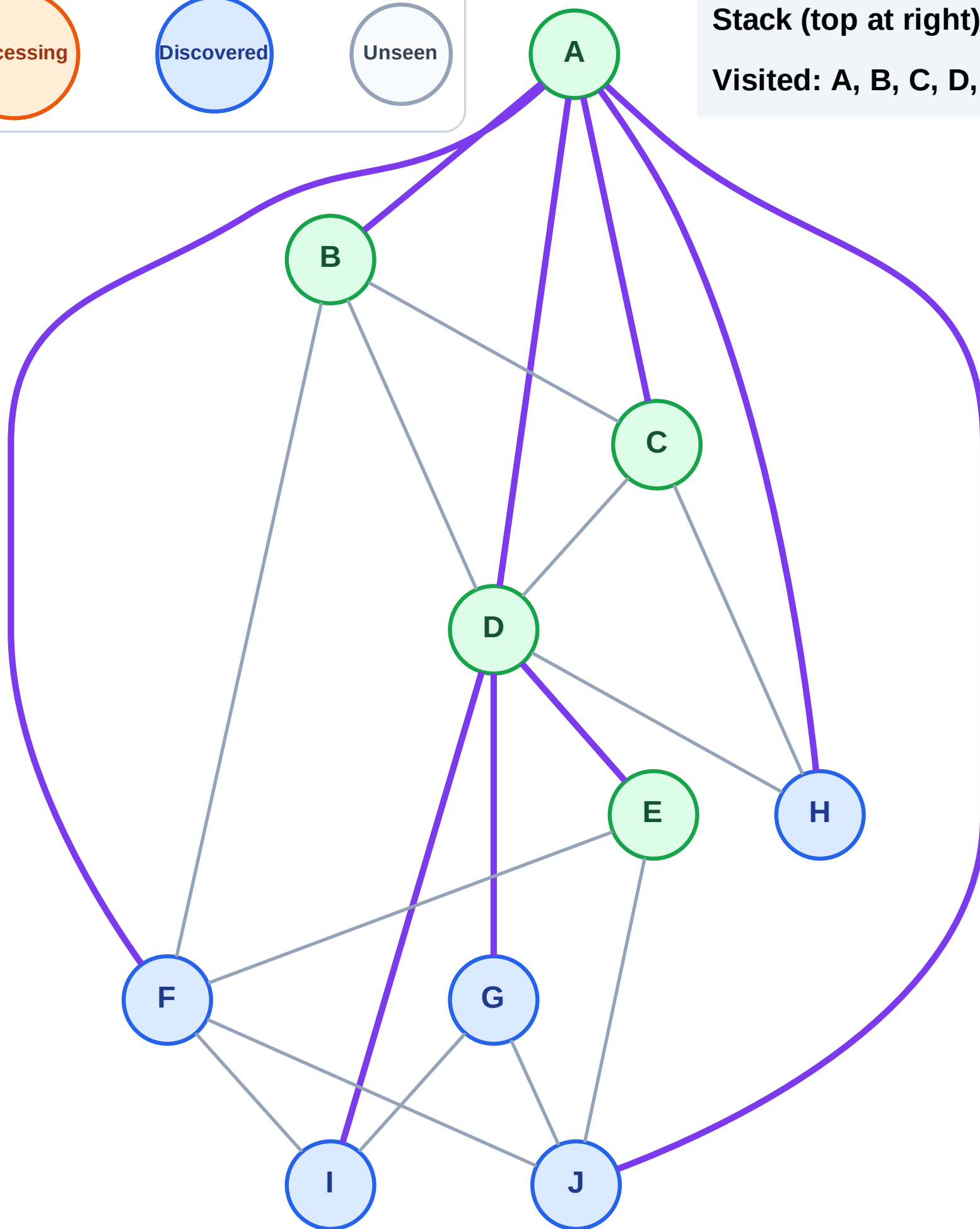
Discovered



Unseen

Stack (top at right): J | H | F | I | G

Visited: A, B, C, D, E



# DFS Traversal

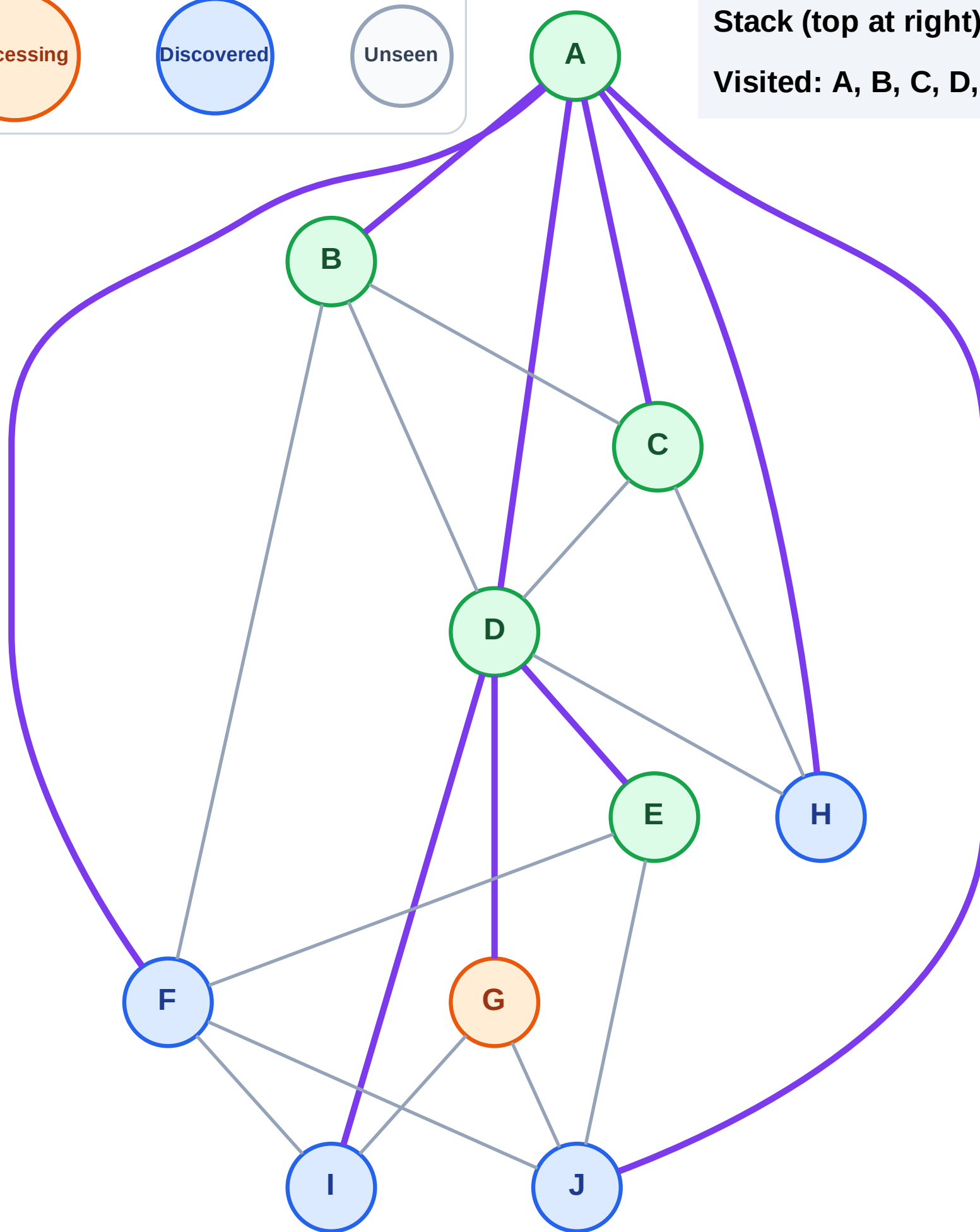
Step 12: Process node G

## Legend



Stack (top at right): J | H | F | I

Visited: A, B, C, D, E



# DFS Traversal

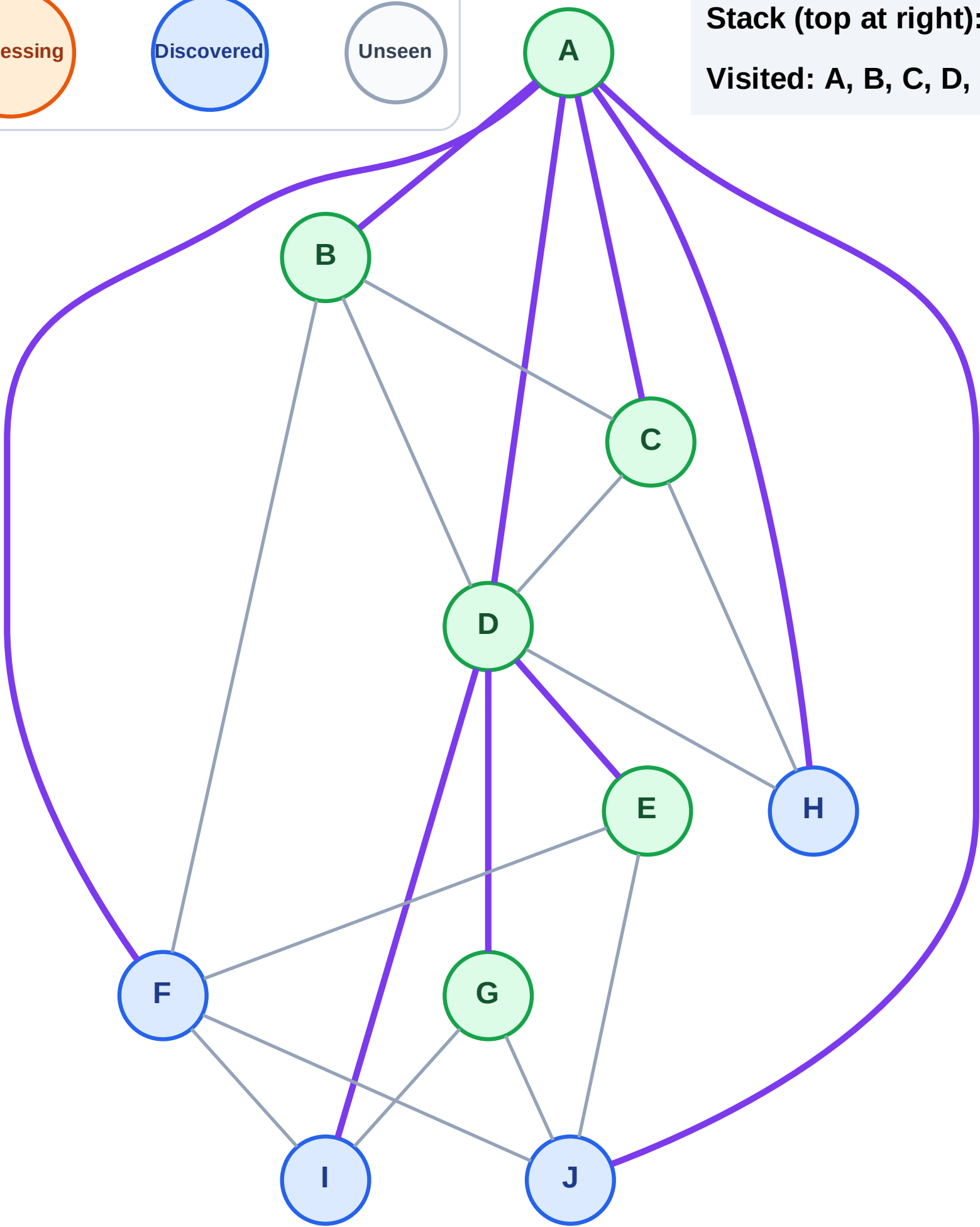
Step 13: No new neighbors from G

## Legend



Stack (top at right): J | H | F | I

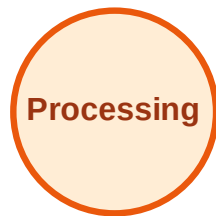
Visited: A, B, C, D, E, G



# DFS Traversal

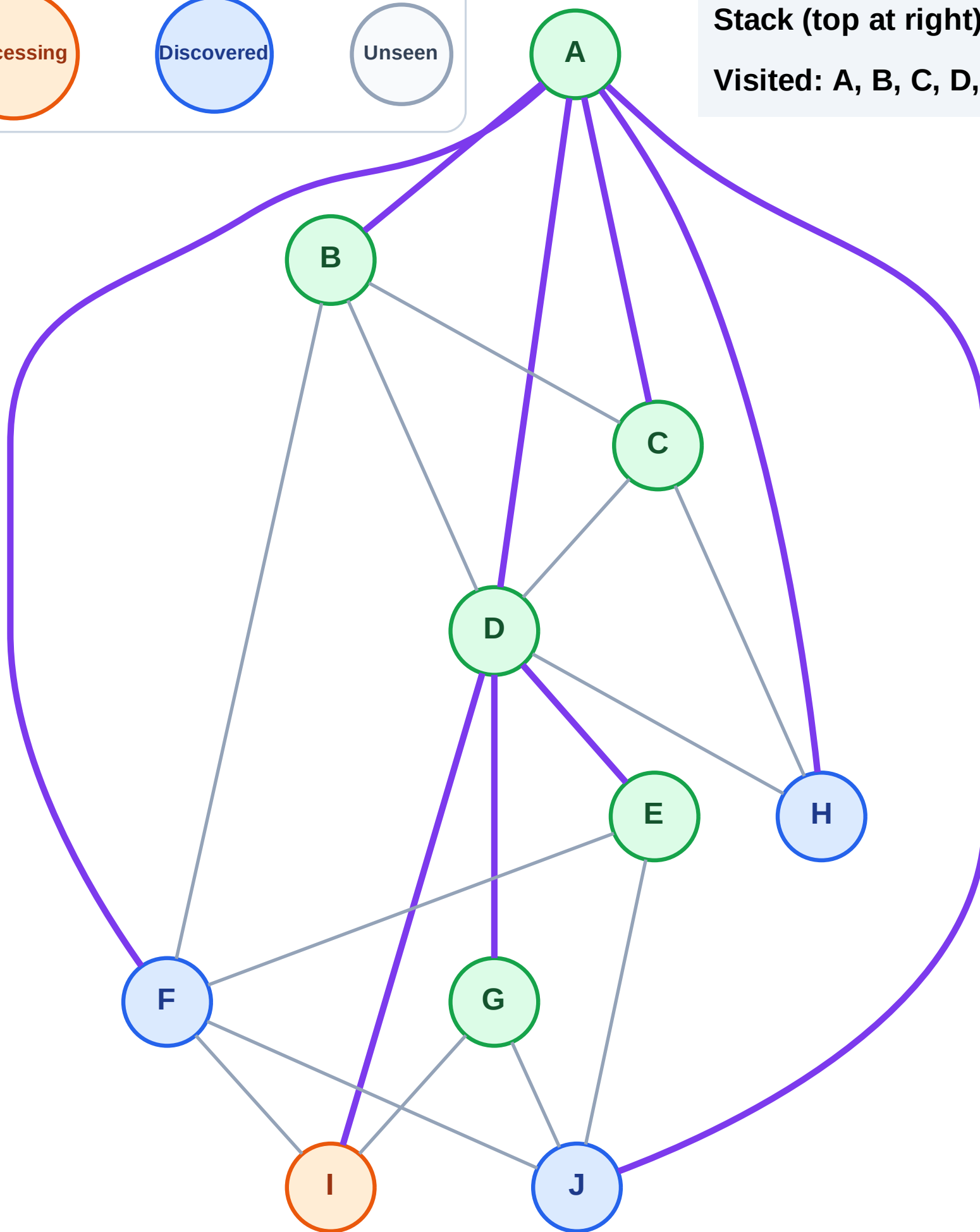
Step 14: Process node I

## Legend



Stack (top at right): J | H | F

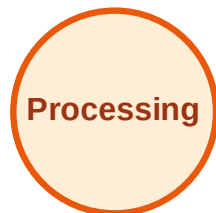
Visited: A, B, C, D, E, G



# DFS Traversal

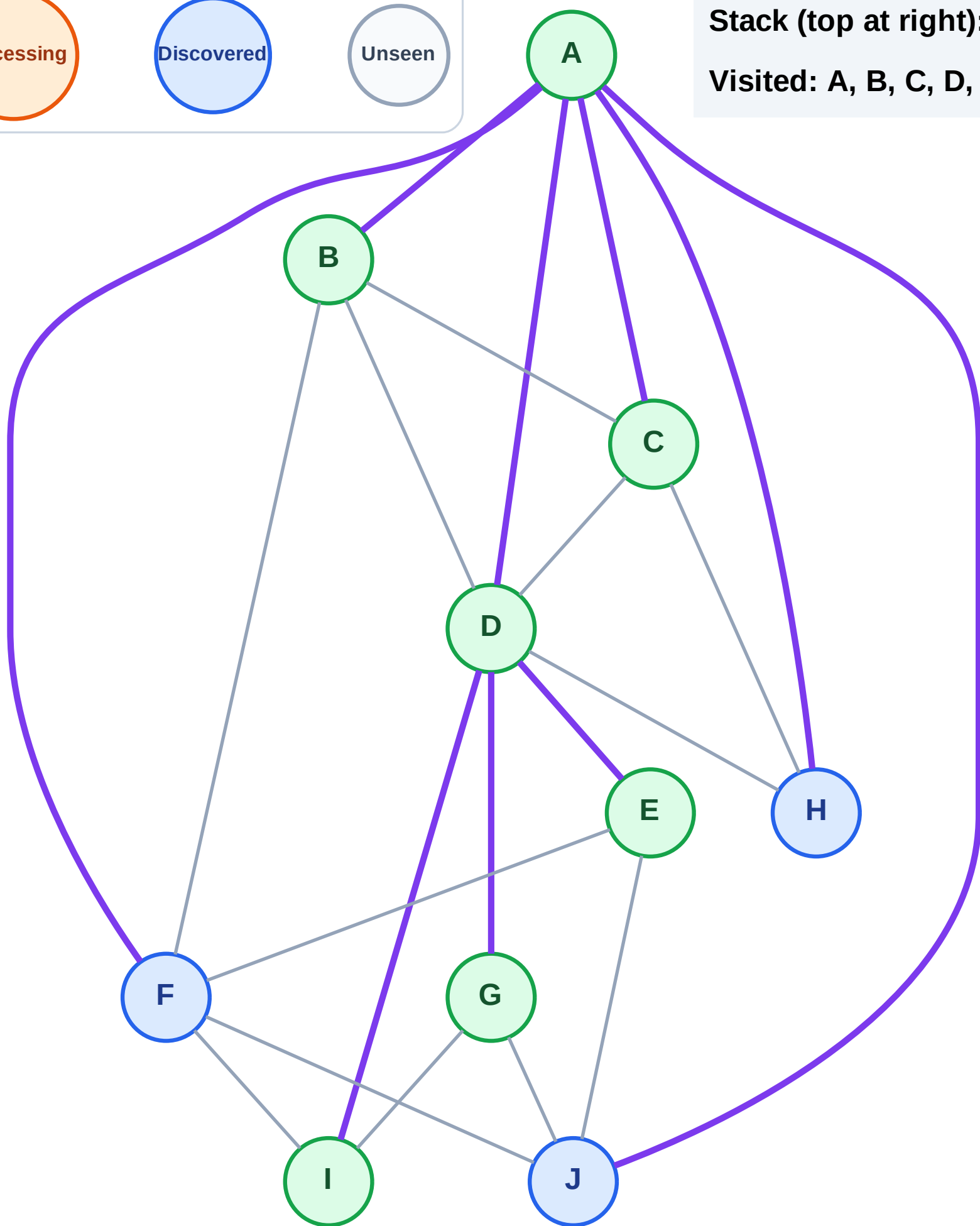
Step 15: No new neighbors from I

## Legend



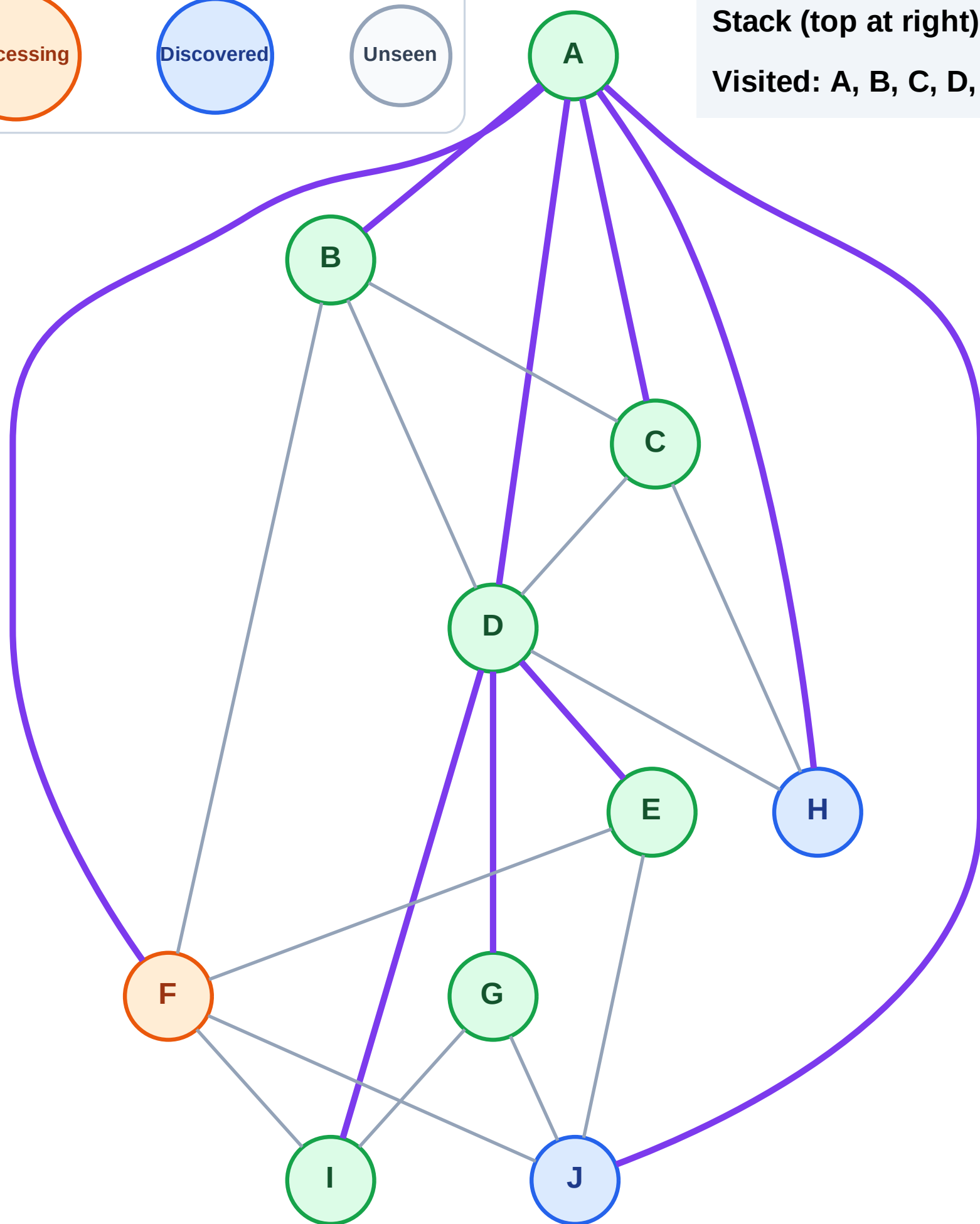
Stack (top at right): J | H | F

Visited: A, B, C, D, E, G, I



### Step 16: Process node F

---



# DFS Traversal

Step 17: No new neighbors from F

## Legend

Complete

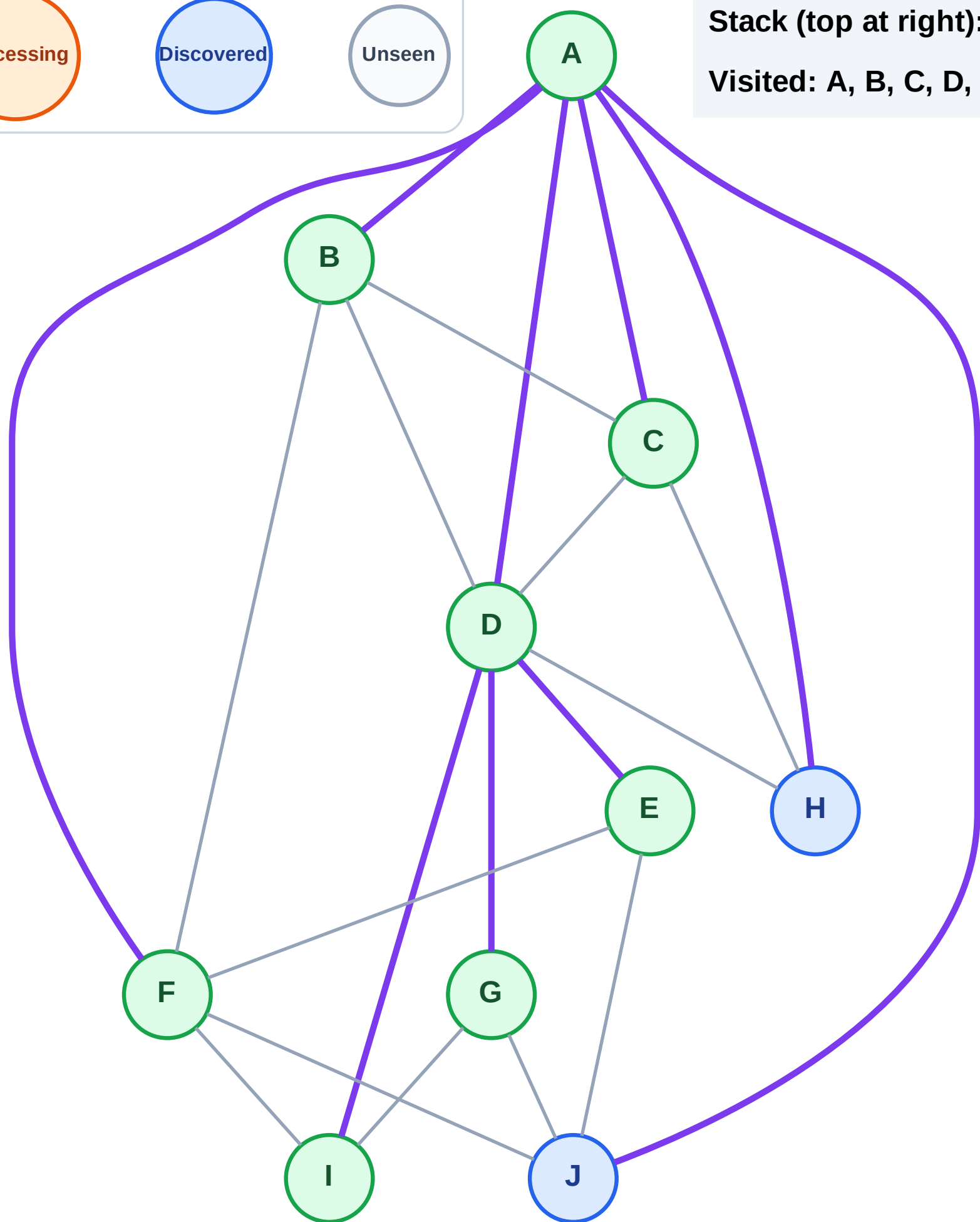
Processing

Discovered

Unseen

Stack (top at right): J | H

Visited: A, B, C, D, E, F, G, I



# DFS Traversal

Step 18: Process node H

## Legend

Complete

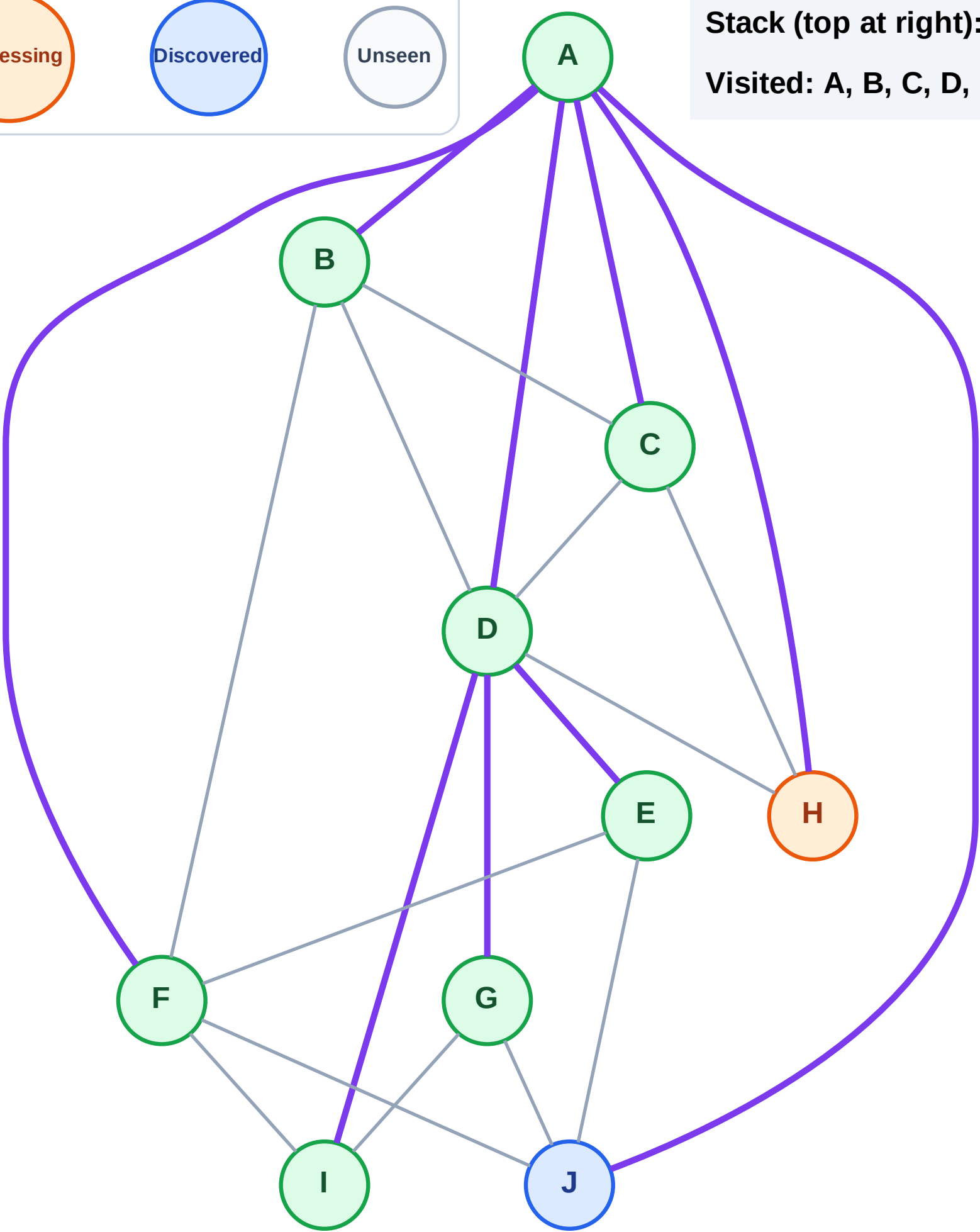
Processing

Discovered

Unseen

Stack (top at right): J

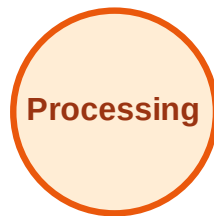
Visited: A, B, C, D, E, F, G, I



# DFS Traversal

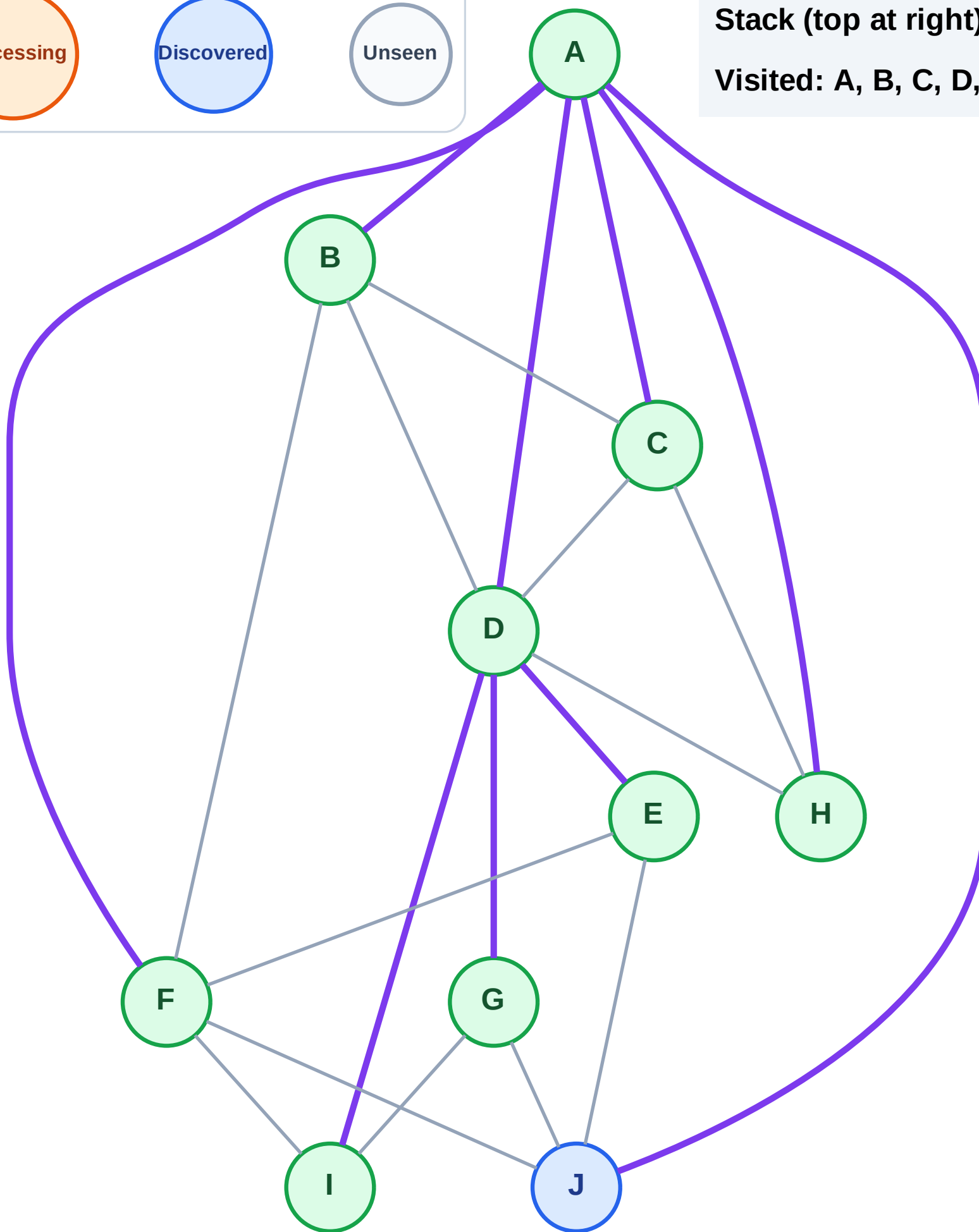
Step 19: No new neighbors from H

## Legend



Stack (top at right): J

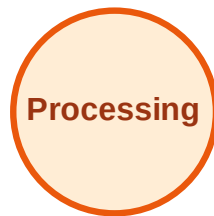
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

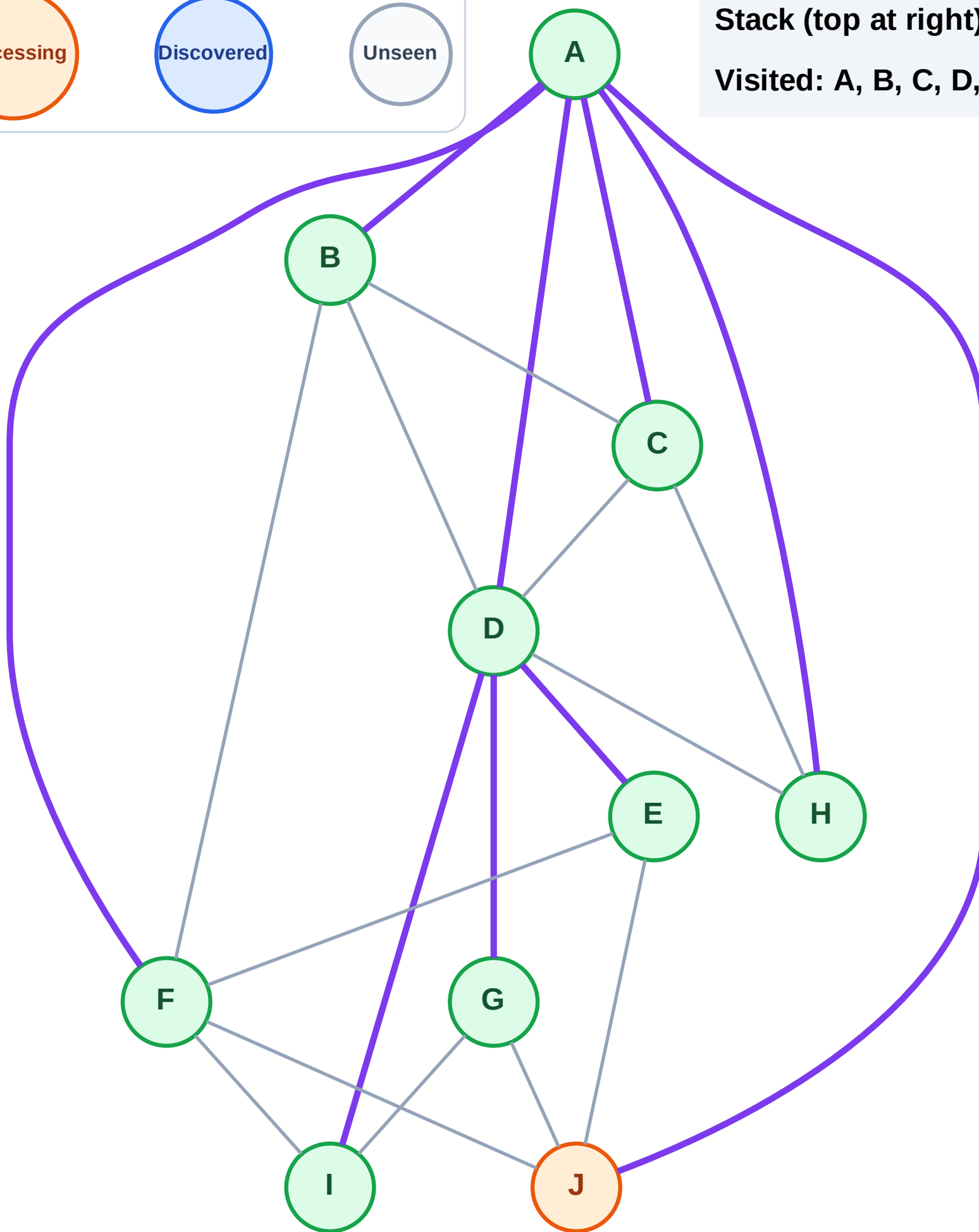
Step 20: Process node J

## Legend



Stack (top at right): (empty)

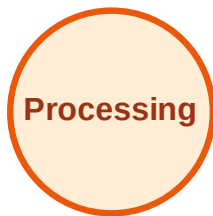
Visited: A, B, C, D, E, F, G, H, I



# DFS Traversal

Step 21: No new neighbors from J

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

